VALIDATION PLAN
OPERATIONAL QUALIFICATION
CLINICAL MODULE

JR Validated Environment — Clinical Module v5.0
Module 1: Study Design | Module 2: Diagnostic Accuracy
Module 3: Survival Analysis | Module 4: Comparative Analysis

Field	Value
Document Number	JR-VP-CLIN-001
Title	Validation Plan — Operational Qualification, Clinical Module
System	JR Validated Environment — Clinical Module
Module Version	5.0
Document Version	5.0 (Rev 5)
Status	Approved
Effective Date	2026-07-20
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Reviewer	[Name]
Approver	[Name]

Version History

Version	Date	Author	Description
1.0	2026-07-15	Joep Rous	Initial release. Covers the 3 clinical study design scripts (Module 1); 36 automated test cases, UR-CLIN-001 to UR-CLIN-036.
2.0	2026-07-16	Joep Rous	Scope extended to Module 2 — Diagnostic Accuracy: jrc_clinical_dx_accuracy, jrc_clinical_dx_ss and jrc_clinical_dx_roc add 51 test cases (UR-CLIN-037 to UR-CLIN-196), for 196 in total. UR-CLIN-001 to UR-CLIN-036 and their test cases are unchanged from Rev 1 and retain their numbering.
3.0	2026-07-20	Joep Rous	Scope extended to Module 3 — Survival Analysis: jrc_clinical_km and jrc_clinical_coxph add 36 test cases (UR-CLIN-088 to UR-CLIN-196), for 196 in total. UR-CLIN-001 to UR-CLIN-087 and their test cases are unchanged from Rev 1 and Rev 2 and retain their numbering.
4.0	2026-07-19	Joep Rous	Module 2 (Diagnostic Accuracy) extended with jrc_clinical_dx_compare (paired AUC comparison, DeLong 1988): 18 test cases (UR-CLIN-124 to UR-CLIN-196), for 196 in total. UR-CLIN-001 to UR-CLIN-123 and their test cases are unchanged from Rev 1 to Rev 3 and retain their numbering.
5.0	2026-07-20	Joep Rous	Scope extended to Module 4 — Comparative Analysis (jrc_clinical_compare_means, jrc_clinical_compare_props, jrc_clinical_ancova, jrc_clinical_logistic): 55 test cases (UR-CLIN-142 to UR-CLIN-196), for 196 in total. UR-CLIN-001 to UR-CLIN-141 and their test cases are unchanged from Rev 1 to Rev 4 and retain their numbering.

Approval Signatures

Role	Name	Signature	Date
Author	Joep Rous		2026-07-20
Reviewer	[Name]		
Approver (QA)	[Name]		

Purpose and Scope

This Operational Qualification (OQ) Validation Plan defines the test cases for the Clinical module of the JR Validated Environment. It specifies the user requirements, test cases, acceptance criteria, and traceability needed to demonstrate that each clinical script performs correctly for its intended use under 21 CFR 820.70(i) and ISO 13485, in support of clinical investigations planned under ISO 14155 / ICH E9 principles.

This plan covers 196 automated test cases executed using the pytest framework via the admin_clinical_oq runner. Test evidence is written to ~/.jrscript/<PROJECT_ID>/validation/clinical_oq_execution_<timestamp>.txt.

Module 1 — Clinical Study Design (Rev 1). Three scripts compute two-arm parallel-group sample sizes: jrc_clinical_ss_means (continuous endpoint), jrc_clinical_ss_props (binary endpoint, risk difference), and jrc_clinical_ss_survival (time-to-event, log-rank). Each supports the applicable hypothesis frameworks (superiority, non-inferiority, and — except survival — equivalence/TOST) via the --framework flag, with allocation ratio, dropout inflation, and sensitivity scenario tables computed in the validated R layer. Statistical methods: Chow, Shao & Wang (2008), Sample Size Calculations in Clinical Research, 2nd ed., Chapters 3 and 4 (normal approximations); Schoenfeld (1983) events formula for the log-rank test. All three scripts use base R only.

Module 2 — Diagnostic Accuracy (added at Rev 2). Three scripts evaluate a diagnostic test against a binary reference standard. jrc_clinical_dx_ss sizes a diagnostic study by target CI half-width or against a performance goal, sizing the reference-positive and reference-negative groups separately and converting to total enrolment via the prevalence (Buderer 1996). jrc_clinical_dx_accuracy reports the 2x2 operating characteristics — sensitivity, specificity, PPV, NPV, LR+, LR- and accuracy — with Clopper-Pearson exact or Wilson intervals, likelihood-ratio intervals on the log scale (Simel 1991), and Bayes-adjusted PPV/NPV at a stated prevalence per FDA (2007), Statistical Guidance on Reporting Results from Studies Evaluating Diagnostic Tests. jrc_clinical_dx_roc reports the empirical ROC curve, the AUC with a DeLong (1988) confidence interval, and the Youden-optimal cutoff. Unlike Module 1, jrc_clinical_dx_accuracy and jrc_clinical_dx_roc consume a data file.

Statistical implementation note. The AUC and its DeLong variance are implemented in base R using the midrank algorithm of Sun & Xu (2014), rather than delegated to an external package; no ROC package is pinned in this environment. Two independent controls apply. First, the acceptance criterion for the AUC (TC-CLIN-DXROC-003) is taken from the original literature — Hanley & McNeil (1982), Radiology 143:29-36, Table 1, which reports an area of 0.893 — rather than from this implementation's own output. Second, jrc_clinical_dx_roc recomputes the AUC at run time by the definitional O(m*n) Mann-Whitney double sum and halts if the two disagree, so every execution re-verifies the algorithm against its definition. jrc_clinical_dx_roc imports ggplot2 to draw its plot; the statistics are base R. The remaining five clinical scripts use base R only.

User Requirements

The following user requirements are tested by this plan.

UR ID	Requirement
UR-CLIN-001	jrc_clinical_ss_means shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-002	jrc_clinical_ss_means shall compute the superiority per-arm and total sample size per the Chow ch. 3 normal-approximation formula.
UR-CLIN-003	jrc_clinical_ss_means shall report the significance level, sidedness, and the resulting z-value.

UR-CLIN-004	jrc_clinical_ss_means shall compute the non-inferiority sample size per the Chow ch. 3 margin formula at one-sided alpha.
UR-CLIN-005	jrc_clinical_ss_means shall inflate the evaluable n to an enrolled N as n/(1−dropout), rounded up per arm.
UR-CLIN-006	jrc_clinical_ss_means shall compute the equivalence sample size with the TOST beta term z(1−(1−power)/2).
UR-CLIN-007	jrc_clinical_ss_means shall support unequal allocation via --ratio (treatment:control).
UR-CLIN-008	jrc_clinical_ss_means shall print an n-vs-SD scenario table over SD × 0.8…1.2 when --sensitivity is given.
UR-CLIN-009	jrc_clinical_ss_means shall exit non-zero when superiority is requested without a meaningful difference.
UR-CLIN-010	jrc_clinical_ss_means shall exit non-zero when non-inferiority is requested without a margin.
UR-CLIN-011	jrc_clinical_ss_means shall exit non-zero when the assumed true difference is not inside the equivalence zone.
UR-CLIN-012	jrc_clinical_ss_means shall exit non-zero when --power is not strictly between 0 and 1.
UR-CLIN-013	jrc_clinical_ss_means shall exit non-zero on any unrecognized argument.
UR-CLIN-014	jrc_clinical_ss_means shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-CLIN-015	jrc_clinical_ss_props shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-016	jrc_clinical_ss_props shall compute the superiority sample size per the Chow ch. 4 unpooled risk-difference formula.
UR-CLIN-017	jrc_clinical_ss_props shall compute the NI sample size, defaulting the treatment rate to the control rate when --p-treat is omitted.
UR-CLIN-018	jrc_clinical_ss_props shall compute the equivalence sample size with the TOST beta term.
UR-CLIN-019	jrc_clinical_ss_props shall exit non-zero when superiority is requested without the expected treatment rate.
UR-CLIN-020	jrc_clinical_ss_props shall exit non-zero when p-treat equals p-control under superiority.
UR-CLIN-021	jrc_clinical_ss_props shall exit non-zero when non-inferiority is requested without a margin.
UR-CLIN-022	jrc_clinical_ss_props shall exit non-zero when --p-control is not strictly between 0 and 1.
UR-CLIN-023	jrc_clinical_ss_props shall print an n-vs-p-control scenario table over p-control × 0.8…1.2 when --sensitivity is given.
UR-CLIN-024	jrc_clinical_ss_props shall inflate the evaluable n to an enrolled N as n/(1−dropout), rounded up per arm.
UR-CLIN-025	jrc_clinical_ss_props shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-CLIN-026	jrc_clinical_ss_survival shall accept valid superiority inputs and exit 0 with the report header present.
UR-CLIN-027	jrc_clinical_ss_survival shall compute required events per Schoenfeld and convert to subjects via the event probability.
UR-CLIN-028	jrc_clinical_ss_survival shall compute NI events using log(margin) − log(true HR) at one-sided alpha.
UR-CLIN-029	jrc_clinical_ss_survival shall exit non-zero when --framework equivalence is requested.
UR-CLIN-030	jrc_clinical_ss_survival shall exit non-zero when superiority is requested with a hazard ratio of exactly 1.
UR-CLIN-031	jrc_clinical_ss_survival shall exit non-zero when the NI margin on the hazard ratio is ≤ 1.
UR-CLIN-032	jrc_clinical_ss_survival shall exit non-zero when the assumed true HR is not less than the margin.
UR-CLIN-033	jrc_clinical_ss_survival shall exit non-zero when the event probability is missing or out of (0, 1].
UR-CLIN-034	jrc_clinical_ss_survival shall print an n-vs-event-prob scenario table over event-prob × 0.8…1.2 when --sensitivity is given.
UR-CLIN-035	jrc_clinical_ss_survival shall inflate the evaluable n to an enrolled N as n/(1−dropout), rounded up per arm.
UR-CLIN-036	jrc_clinical_ss_survival shall exit non-zero and report RENV_PATHS_ROOT when called directly via Rscript without the wrapper.
UR-CLIN-037	jrc_clinical_dx_accuracy shall accept a valid id/reference/result CSV and exit 0 with the report header present.
UR-CLIN-038	jrc_clinical_dx_accuracy shall cross-tabulate the index test against the reference standard and report TP, FP, FN, TN with their margins.
UR-CLIN-039	jrc_clinical_dx_accuracy shall compute sensitivity as TP/(TP+FN) with a two-sided Clopper-Pearson exact confidence interval.

UR-CLIN-040	jrc_clinical_dx_accuracy shall compute specificity as TN/(TN+FP) with a two-sided Clopper-Pearson exact confidence interval.
UR-CLIN-041	jrc_clinical_dx_accuracy shall compute overall accuracy as (TP+TN)/N with a two-sided exact confidence interval.
UR-CLIN-042	jrc_clinical_dx_accuracy shall compute LR+ as sens/(1-spec) with the log (Simel 1991) confidence interval.
UR-CLIN-043	jrc_clinical_dx_accuracy shall compute LR- as (1-sens)/spec with the log (Simel 1991) confidence interval.
UR-CLIN-044	jrc_clinical_dx_accuracy shall report PPV and NPV computed from the study's own row totals, state the study prevalence used, and warn that they are valid only if it reflects the intended-use population.
UR-CLIN-045	jrc_clinical_dx_accuracy shall offer a Wilson score interval via --ci wilson, leaving the point estimate unchanged.
UR-CLIN-046	jrc_clinical_dx_accuracy shall, given --prevalence P, additionally report PPV and NPV recomputed by Bayes' theorem at P.
UR-CLIN-047	jrc_clinical_dx_accuracy shall honour --conf, producing a strictly narrower interval at a lower confidence level.
UR-CLIN-048	jrc_clinical_dx_accuracy shall accept arbitrary category labels when the positive label is named with --positive.
UR-CLIN-049	jrc_clinical_dx_accuracy shall drop rows with a missing id, reference or result, and report how many were dropped.
UR-CLIN-050	jrc_clinical_dx_accuracy shall reject an unrecognised --ci method.
UR-CLIN-051	jrc_clinical_dx_accuracy shall reject a --prevalence outside (0, 1).
UR-CLIN-052	jrc_clinical_dx_accuracy shall reject a CSV lacking a required column and name the missing column.
UR-CLIN-053	jrc_clinical_dx_accuracy shall reject a CSV containing duplicate ids.
UR-CLIN-054	jrc_clinical_dx_accuracy shall reject category labels it cannot map to positive/negative, and say so.
UR-CLIN-055	jrc_clinical_dx_accuracy shall refuse to report when the reference standard has no negative subjects.
UR-CLIN-056	jrc_clinical_dx_accuracy shall refuse to run outside the validated environment (RENV_PATHS_ROOT unset).
UR-CLIN-057	jrc_clinical_dx_ss shall accept valid precision-method inputs and exit 0 with the report header present.
UR-CLIN-058	jrc_clinical_dx_ss shall size each reference arm for a target CI half-width and convert to total enrolment via the prevalence, per Buderer (1996).
UR-CLIN-059	jrc_clinical_dx_ss shall size each reference arm to reject a performance goal at the stated power, and convert to total enrolment.
UR-CLIN-060	jrc_clinical_dx_ss shall report which reference arm drives the total enrolment.
UR-CLIN-061	jrc_clinical_dx_ss shall leave the per-arm requirements unchanged when the prevalence changes, while raising the total enrolment.
UR-CLIN-062	jrc_clinical_dx_ss shall, given --sensitivity, print total N across a range of prevalence values with the assumed value marked.
UR-CLIN-063	jrc_clinical_dx_ss shall inflate the evaluable N to an enrolled N for the stated unevaluable fraction.
UR-CLIN-064	jrc_clinical_dx_ss shall default to a two-sided z for the precision method and a one-sided z for the hypothesis method.
UR-CLIN-065	jrc_clinical_dx_ss shall reject the precision method without --halfwidth.
UR-CLIN-066	jrc_clinical_dx_ss shall reject the hypothesis method without --sens-goal.
UR-CLIN-067	jrc_clinical_dx_ss shall reject a performance goal at or above the expected value.
UR-CLIN-068	jrc_clinical_dx_ss shall reject a --prevalence outside (0, 1).
UR-CLIN-069	jrc_clinical_dx_ss shall reject an unrecognised --method.
UR-CLIN-070	jrc_clinical_dx_ss shall refuse to run outside the validated environment (RENV_PATHS_ROOT unset).
UR-CLIN-071	jrc_clinical_dx_roc shall accept a valid id/reference/score CSV and exit 0 with the report header present.
UR-CLIN-072	jrc_clinical_dx_roc shall save a two-panel PNG to ~/Downloads/.
UR-CLIN-073	jrc_clinical_dx_roc shall reproduce the area under the ROC curve published by Hanley & McNeil (1982) for their Table 1 rating-scale data.
UR-CLIN-074	jrc_clinical_dx_roc shall report the AUC with its DeLong standard error and a two-sided confidence interval.
UR-CLIN-075	jrc_clinical_dx_roc shall score tied scores as 0.5 in the Mann-Whitney kernel.
UR-CLIN-076	jrc_clinical_dx_roc shall report the cutoff maximising Youden's J = sensitivity + specificity - 1, with J

	and the operating point.
UR-CLIN-077	jrc_clinical_dx_roc shall report the cutoff as an observed score value together with the explicit rule that applies it.
UR-CLIN-078	jrc_clinical_dx_roc shall test the AUC against 0.5 using the DeLong standard error and report z and p.
UR-CLIN-079	jrc_clinical_dx_roc shall honour --direction lower by reflecting the AUC about 0.5.
UR-CLIN-080	jrc_clinical_dx_roc shall report that no cutoff exists, rather than an infinite threshold, when no candidate cutoff beats chance in the stated direction; and shall warn when AUC < 0.5.
UR-CLIN-081	jrc_clinical_dx_roc shall offer a logit-transformed AUC interval via --ci-method logit, leaving the point estimate unchanged.
UR-CLIN-082	jrc_clinical_dx_roc shall honour --conf, producing a strictly narrower interval at a lower confidence level.
UR-CLIN-083	jrc_clinical_dx_roc shall reject an unrecognised --direction.
UR-CLIN-084	jrc_clinical_dx_roc shall reject a CSV lacking a required column and name the missing column.
UR-CLIN-085	jrc_clinical_dx_roc shall reject a CSV containing duplicate ids.
UR-CLIN-086	jrc_clinical_dx_roc shall refuse to run when the reference standard has fewer than two negative subjects.
UR-CLIN-087	jrc_clinical_dx_roc shall refuse to run outside the validated environment (RENV_PATHS_ROOT unset).
UR-CLIN-088	jrc_clinical_km shall accept a valid id/time/event CSV and exit 0 with the report header present.
UR-CLIN-089	jrc_clinical_km shall save a two-panel PNG (curves + number-at-risk) to the output directory.
UR-CLIN-090	jrc_clinical_km shall reproduce the median survival times published for the Miller (1981) acute-myelogenous-leukaemia maintenance trial.
UR-CLIN-091	jrc_clinical_km shall reproduce the log-rank statistic for the Miller (1981) aml comparison.
UR-CLIN-092	jrc_clinical_km shall report observed and expected events per group.
UR-CLIN-093	jrc_clinical_km shall report the overall median survival with a confidence interval when no group column is given.
UR-CLIN-094	jrc_clinical_km shall report S(t) with a confidence interval at a requested time via --time-point.
UR-CLIN-095	jrc_clinical_km shall apply --conf to the reported intervals.
UR-CLIN-096	jrc_clinical_km shall compute the rho-family weighted test when --rho is given.
UR-CLIN-097	jrc_clinical_km shall accept --event-positive to name the event label.
UR-CLIN-098	jrc_clinical_km shall auto-detect common event label schemes.
UR-CLIN-099	jrc_clinical_km shall report a median-CI bound as NA when the curve does not reach it within follow-up.
UR-CLIN-100	jrc_clinical_km shall reject a CSV missing id, time or event.
UR-CLIN-101	jrc_clinical_km shall reject a CSV with a duplicated id.
UR-CLIN-102	jrc_clinical_km shall reject data in which no event is observed.
UR-CLIN-103	jrc_clinical_km shall reject --group when the column has one value.
UR-CLIN-104	jrc_clinical_km shall reject an event column whose labels are not recognised and not named via --event-positive.
UR-CLIN-105	jrc_clinical_km shall reject a non-positive time value.
UR-CLIN-106	jrc_clinical_km shall refuse to run outside the validated environment.
UR-CLIN-107	jrc_clinical_coxph shall accept a valid id/time/event CSV with covariates and exit 0 with the report header present.
UR-CLIN-108	jrc_clinical_coxph shall save a Schoenfeld-residual diagnostic PNG to the output directory.
UR-CLIN-109	jrc_clinical_coxph shall reproduce the sex hazard ratio published for the NCCTG lung-cancer Cox model.
UR-CLIN-110	jrc_clinical_coxph shall reproduce the age hazard ratio for the same lung model.
UR-CLIN-111	jrc_clinical_coxph shall report Harrell's concordance for the model.
UR-CLIN-112	jrc_clinical_coxph shall report the global likelihood-ratio and Wald chi-square tests.
UR-CLIN-113	jrc_clinical_coxph shall test the proportional-hazards assumption and report it as satisfied when it holds.
UR-CLIN-114	jrc_clinical_coxph shall flag a proportional-hazards violation on the Veterans' lung-cancer Karnofsky model.
UR-CLIN-115	jrc_clinical_coxph shall treat a text covariate as a factor with the first level as reference.
UR-CLIN-116	jrc_clinical_coxph shall drop rows with a missing value in any named covariate before fitting.
UR-CLIN-117	jrc_clinical_coxph shall apply the tie method selected by --ties.
UR-CLIN-118	jrc_clinical_coxph shall apply --conf to the hazard-ratio intervals.
UR-CLIN-119	jrc_clinical_coxph shall reject a --covariates column not present in the data.

UR-CLIN-120	jrc_clinical_coxph shall require at least one covariate.
UR-CLIN-121	jrc_clinical_coxph shall reject a CSV with a duplicated id.
UR-CLIN-122	jrc_clinical_coxph shall refuse to fit with fewer than two events.
UR-CLIN-123	jrc_clinical_coxph shall refuse to run outside the validated environment.
UR-CLIN-124	jrc_clinical_dx_compare shall accept a valid id/reference + two-score CSV and exit 0 with the report header present.
UR-CLIN-125	jrc_clinical_dx_compare shall save a two-panel PNG to the output directory.
UR-CLIN-126	jrc_clinical_dx_compare shall reproduce the two AUCs returned by the reference implementation pROC on the seed-7 fixture.
UR-CLIN-127	jrc_clinical_dx_compare shall report the difference A - B with a confidence interval.
UR-CLIN-128	jrc_clinical_dx_compare shall report the paired DeLong z statistic and p value, matching pROC.
UR-CLIN-129	jrc_clinical_dx_compare shall report the correlation between the two AUCs.
UR-CLIN-130	jrc_clinical_dx_compare shall state which test has the higher AUC and whether the difference is significant.
UR-CLIN-131	jrc_clinical_dx_compare shall apply --conf to the difference CI.
UR-CLIN-132	jrc_clinical_dx_compare shall build per-test AUC CIs on the logit scale under --ci-method logit.
UR-CLIN-133	jrc_clinical_dx_compare shall apply --direction to both tests.
UR-CLIN-134	jrc_clinical_dx_compare shall accept --tests to name the two score columns.
UR-CLIN-135	jrc_clinical_dx_compare shall infer the two test columns when the file has exactly two besides id and reference.
UR-CLIN-136	jrc_clinical_dx_compare shall reject a file with more than two candidate test columns and no --tests.
UR-CLIN-137	jrc_clinical_dx_compare shall reject a --tests column not in the data.
UR-CLIN-138	jrc_clinical_dx_compare shall reject a CSV without a reference column.
UR-CLIN-139	jrc_clinical_dx_compare shall reject a CSV with a duplicated id.
UR-CLIN-140	jrc_clinical_dx_compare shall reject data with fewer than two reference-negative subjects.
UR-CLIN-141	jrc_clinical_dx_compare shall refuse to run outside the validated environment.
UR-CLIN-142	jrc_clinical_compare_means shall accept a valid id/group/value CSV and exit 0 with the report header present.
UR-CLIN-143	jrc_clinical_compare_means shall reproduce the Welch two-sample t-test on Student's (1908) extra-sleep data.
UR-CLIN-144	jrc_clinical_compare_means shall report the mean difference with a CI.
UR-CLIN-145	jrc_clinical_compare_means shall report Cohen's d and Hedges' g.
UR-CLIN-146	jrc_clinical_compare_means shall report the Wilcoxon rank-sum test.
UR-CLIN-147	jrc_clinical_compare_means shall use the equal-variance t-test with --test student.
UR-CLIN-148	jrc_clinical_compare_means shall apply --conf to the interval.
UR-CLIN-149	jrc_clinical_compare_means shall report a Shapiro-Wilk p per group.
UR-CLIN-150	jrc_clinical_compare_means shall compute one-way ANOVA for >2 groups.
UR-CLIN-151	jrc_clinical_compare_means shall reject a CSV missing id/group/value.
UR-CLIN-152	jrc_clinical_compare_means shall reject a duplicated id.
UR-CLIN-153	jrc_clinical_compare_means shall reject fewer than two groups.
UR-CLIN-154	jrc_clinical_compare_means shall reject a group with < 2 observations.
UR-CLIN-155	jrc_clinical_compare_means shall reject an unknown --test value.
UR-CLIN-156	jrc_clinical_compare_means shall refuse to run outside the validated env.
UR-CLIN-157	jrc_clinical_compare_props shall accept a valid id/group/outcome CSV and exit 0 with the contingency table.
UR-CLIN-158	jrc_clinical_compare_props shall report the Yates-corrected chi-square.
UR-CLIN-159	jrc_clinical_compare_props shall report Fisher's exact p.
UR-CLIN-160	jrc_clinical_compare_props shall report the risk difference with a CI.
UR-CLIN-161	jrc_clinical_compare_props shall report the risk ratio with a CI.
UR-CLIN-162	jrc_clinical_compare_props shall report the odds ratio with a CI.
UR-CLIN-163	jrc_clinical_compare_props shall drop the Yates correction with --no-correction.
UR-CLIN-164	jrc_clinical_compare_props shall honour --reference for the odds ratio.
UR-CLIN-165	jrc_clinical_compare_props shall auto-detect the event outcome level.
UR-CLIN-166	jrc_clinical_compare_props shall reject a CSV missing a required column.
UR-CLIN-167	jrc_clinical_compare_props shall reject a duplicated id.
UR-CLIN-168	jrc_clinical_compare_props shall reject fewer than two groups.

UR-CLIN-169	jrc_clinical_compare_props shall reject an --event not in the data.
UR-CLIN-170	jrc_clinical_compare_props shall refuse to run outside the validated env.
UR-CLIN-171	jrc_clinical_ancova shall accept a valid id/group/value + covariate CSV and exit 0 with the header.
UR-CLIN-172	jrc_clinical_ancova shall reproduce the covariate-adjusted group test for the anorexia data.
UR-CLIN-173	jrc_clinical_ancova shall report the Type III test for the covariate.
UR-CLIN-174	jrc_clinical_ancova shall report the adjusted (LS) means per group.
UR-CLIN-175	jrc_clinical_ancova shall report multiplicity-adjusted pairwise contrasts.
UR-CLIN-176	jrc_clinical_ancova shall report each covariate's coefficient.
UR-CLIN-177	jrc_clinical_ancova shall apply --conf to the adjusted-mean CIs.
UR-CLIN-178	jrc_clinical_ancova shall reject a CSV missing id/group/value.
UR-CLIN-179	jrc_clinical_ancova shall require at least one covariate.
UR-CLIN-180	jrc_clinical_ancova shall reject a covariate not in the data.
UR-CLIN-181	jrc_clinical_ancova shall reject a duplicated id.
UR-CLIN-182	jrc_clinical_ancova shall reject fewer than two group levels.
UR-CLIN-183	jrc_clinical_ancova shall refuse to run outside the validated env.
UR-CLIN-184	jrc_clinical_logistic shall accept a valid id/outcome + predictor CSV and exit 0 with the header.
UR-CLIN-185	jrc_clinical_logistic shall reproduce the smoking odds ratio for the low-birth-weight data.
UR-CLIN-186	jrc_clinical_logistic shall report the odds ratio for a continuous predictor.
UR-CLIN-187	jrc_clinical_logistic shall report a non-significant predictor faithfully.
UR-CLIN-188	jrc_clinical_logistic shall report the LR test against the null model.
UR-CLIN-189	jrc_clinical_logistic shall report the model AIC.
UR-CLIN-190	jrc_clinical_logistic shall report the AUC of the fitted probabilities.
UR-CLIN-191	jrc_clinical_logistic shall apply --conf to the odds-ratio CIs.
UR-CLIN-192	jrc_clinical_logistic shall reject a CSV missing id/outcome/predictor.
UR-CLIN-193	jrc_clinical_logistic shall require at least one predictor.
UR-CLIN-194	jrc_clinical_logistic shall reject an outcome without exactly two levels.
UR-CLIN-195	jrc_clinical_logistic shall reject a duplicated id.
UR-CLIN-196	jrc_clinical_logistic shall refuse to run outside the validated env.

Test Environment

All test cases are executed via the admin_clinical_oq runner using pytest with the shared OQ virtual environment at ~/.venvs/<PROJECT_ID>_oq/. Evidence is written to ~/.jrscript/<PROJECT_ID>/validation/.

The four sample-size scripts (jrc_clinical_ss_means, jrc_clinical_ss_props, jrc_clinical_ss_survival, jrc_clinical_dx_ss) are parameter-only: study design happens before data exist, so their test cases pass parameters on the command line and assert on the report. The two analysis scripts added at Rev 2 (jrc_clinical_dx_accuracy, jrc_clinical_dx_roc) consume a CSV; their fixtures are fixed, committed files in repos/clinical/oq/data/, so every execution of the suite is reproducible.

Item	Specification
Language	R (≥ 4.3.0 recommended)
R packages	base R only, except jrc_clinical_dx_roc, which imports ggplot2 (pinned in admin/R_requirements.txt) to draw its plot. All statistics in all six scripts are base R.
Test framework	pytest (≥ 8.0)
Test runner	admin_clinical_oq
OQ venv	~/.venvs/<PROJECT_ID>_oq/ (shared with admin_oq)
Test data	repos/clinical/oq/data/ — fixed committed CSV fixtures (Rev 2; the Rev 1 suite used none)
Evidence output	~/.jrscript/<PROJECT_ID>/validation/clinical_oq_execution_<timestamp>.txt
Validation plan ID	JR-VP-CLIN-001 Rev 2

Test Cases — jrc_clinical_ss_means

TC-CLIN-MEANS-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-001
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5`
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'continuous endpoint'.

TC-CLIN-MEANS-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-002
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5`
Rationale	Verifies the core superiority formula against the independently derived reference: n/arm = ceiling($2 \cdot 10^2 \cdot (z_{0.975}+z_{0.80})^2/5^2$) = 63.
Pass criterion	n treatment = 63, n control = 63, n TOTAL = 126.

TC-CLIN-MEANS-003 Two-sided z-value reported

Tests UR(s)	UR-CLIN-003
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5`
Rationale	The sidedness convention is the guide's #1 pitfall; the report must state it and the z it implies.
Pass criterion	Output contains '2-sided' and z = 1.9600.

TC-CLIN-MEANS-004 Non-inferiority numeric correctness

Tests UR(s)	UR-CLIN-004
Command / action	`jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8 --margin 4`
Rationale	Verifies the NI branch: n/arm = ceiling($2 \cdot 8^2 \cdot (z_{0.975}+z_{0.90})^2/4^2$) = 85 (alpha 0.025 one-sided).
Pass criterion	n treatment = 85, n control = 85, n TOTAL = 170.

TC-CLIN-MEANS-005 Dropout inflation

Tests UR(s)	UR-CLIN-005
Command / action	`jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8 --margin 4 --dropout 0.15`
Rationale	Enrolled N is a script responsibility (never GUI arithmetic): ceiling(85/0.85) = 100 per arm.
Pass criterion	Output contains 'ENROLL 100 + 100 = 200'.

TC-CLIN-MEANS-006 Equivalence (TOST) numeric correctness

Tests UR(s)	UR-CLIN-006
Command / action	`jrc_clinical_ss_means --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --sd 10 --margin 5`
Rationale	Verifies the TOST branch: n/arm = ceiling($2 \cdot 10^2 \cdot (z_{0.95}+z_{0.90})^2/5^2$) = 69.
Pass criterion	n TOTAL = 138. Output contains 'equivalence (TOST)'.

TC-CLIN-MEANS-007 Allocation ratio 2:1

Tests UR(s)	UR-CLIN-007
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --ratio 2`
Rationale	Verifies the (1+1/k) allocation factor: control = ceiling($1.5 \cdot 100 \cdot 2.801585^2/25$) = 48; treatment = 95.
Pass criterion	n treatment = 95, n control = 48.

TC-CLIN-MEANS-008 Sensitivity table scales with SD²

Tests UR(s)	UR-CLIN-008
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --sensitivity`
Rationale	The SD is the most uncertain design input; the table must recompute n per scenario in the

	validated layer ($n \propto SD^2$).
Pass criterion	5 scenario rows; totals 82 at SD=8 and 182 at SD=12; assumed row marked.

TC-CLIN-MEANS-009 Superiority requires --delta

Tests UR(s)	UR-CLIN-009
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10`
Rationale	A superiority design without delta is meaningless; the script must refuse rather than assume.
Pass criterion	Exit code ≠ 0. 'requires --delta' in combined stdout/stderr.

TC-CLIN-MEANS-010 Non-inferiority requires --margin

Tests UR(s)	UR-CLIN-010
Command / action	`jrc_clinical_ss_means --framework non_inferiority --power 0.90 --alpha 0.025 --sides 1 --sd 8`
Rationale	The NI margin is the defining design input; it must be explicit.
Pass criterion	Exit code ≠ 0. 'requires --margin' in combined stdout/stderr.

TC-CLIN-MEANS-011 Equivalence needs |delta| < margin

Tests UR(s)	UR-CLIN-011
Command / action	`jrc_clinical_ss_means --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --sd 10 --margin 5 --delta 5`
Rationale	|delta| ≥ margin makes the required n infinite; the script must refuse with a clear message.
Pass criterion	Exit code ≠ 0. '|delta| < margin' in combined stdout/stderr.

TC-CLIN-MEANS-012 --power out of range

Tests UR(s)	UR-CLIN-012
Command / action	`jrc_clinical_ss_means --framework superiority --power 1.5 --alpha 0.05 --sides 2 --sd 10 --delta 5`
Rationale	Guards against a mistyped power silently producing nonsense.
Pass criterion	Exit code ≠ 0. '--power' in combined stdout/stderr.

TC-CLIN-MEANS-013 Unknown flag rejected

Tests UR(s)	UR-CLIN-013
Command / action	`jrc_clinical_ss_means --framework superiority --power 0.80 --alpha 0.05 --sides 2 --sd 10 --delta 5 --bogus 1`
Rationale	A typo in a flag name must never be silently ignored in a validated tool.
Pass criterion	Exit code ≠ 0. 'Unknown argument' in combined stdout/stderr.

TC-CLIN-MEANS-014 Bypass protection

Tests UR(s)	UR-CLIN-014
Command / action	`Rscript repos/clinical/R/jrc_clinical_ss_means.R ... (RENV_PATHS_ROOT unset)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_ss_props

TC-CLIN-PROPS-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-015
Command / action	`jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.85`
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'binary endpoint'.

TC-CLIN-PROPS-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-016
Command / action	`jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.85`
Rationale	Verifies the core formula: V = 0.85·0.15 + 0.70·0.30 = 0.3375; n/arm = ceiling(0.3375·(z0.975+z0.80)²/0.15²) = 118.
Pass criterion	n treatment = 118, n control = 118, n TOTAL = 236.

TC-CLIN-PROPS-003 Non-inferiority numeric, default p-treat

Tests UR(s)	UR-CLIN-017
Command / action	`jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85 --margin 0.10`
Rationale	Verifies the NI branch and the documented default: V = 2·0.85·0.15 = 0.255; n/arm = ceiling(0.255·7.848886/0.01) = 201.
Pass criterion	n/arm = 201, n TOTAL = 402; 'assumed equal to control' in output.

TC-CLIN-PROPS-004 Equivalence (TOST) numeric correctness

Tests UR(s)	UR-CLIN-018
Command / action	`jrc_clinical_ss_props --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --p-control 0.80 --margin 0.15`
Rationale	Verifies the TOST branch: V = 2·0.80·0.20 = 0.32; n/arm = ceiling(0.32·(z0.95+z0.90)²/0.15²) = 122.
Pass criterion	n TOTAL = 244. Output contains 'equivalence (TOST)'.

TC-CLIN-PROPS-005 Superiority requires --p-treat

Tests UR(s)	UR-CLIN-019
Command / action	`jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70`
Rationale	A superiority design needs the treatment rate; the NI/equivalence default must not leak into superiority.
Pass criterion	Exit code ≠ 0. 'requires --p-treat' in combined stdout/stderr.

TC-CLIN-PROPS-006 Equal rates rejected for superiority

Tests UR(s)	UR-CLIN-020
Command / action	`jrc_clinical_ss_props --framework superiority --power 0.80 --alpha 0.05 --sides 2 --p-control 0.70 --p-treat 0.70`
Rationale	Zero effect size gives an infinite n; the script must refuse.
Pass criterion	Exit code ≠ 0. 'different from p-control' in combined stdout/stderr.

TC-CLIN-PROPS-007 Non-inferiority requires --margin

Tests UR(s)	UR-CLIN-021
Command / action	`jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85`
Rationale	The NI margin is the defining design input; it must be explicit.
Pass criterion	Exit code ≠ 0. 'requires --margin' in combined stdout/stderr.

TC-CLIN-PROPS-008 --p-control out of range

Tests UR(s)	UR-CLIN-022
Command / action	`jrc_clinical_ss_props --framework non_inferiority --power 0.80 --`

	`alpha 0.025 --sides 1 --p-control 1.2 --margin 0.10`
Rationale	Guards against an impossible rate silently producing nonsense.
Pass criterion	Exit code ≠ 0. '--p-control' in combined stdout/stderr.

TC-CLIN-PROPS-009 Sensitivity table over the control rate

Tests UR(s)	UR-CLIN-023
Command / action	`jrc_clinical_ss_props --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --p-control 0.80 --margin 0.15 --sensitivity`
Rationale	The control rate is the most uncertain design input; scenarios must be recomputed in the validated layer.
Pass criterion	5 scenario rows (0.64…0.96); assumed row marked.

TC-CLIN-PROPS-010 Dropout inflation

Tests UR(s)	UR-CLIN-024
Command / action	`jrc_clinical_ss_props --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --p-control 0.85 --margin 0.10 --dropout 0.10`
Rationale	Enrolled N is a script responsibility: ceiling(201/0.90) = 224 per arm.
Pass criterion	Output contains 'ENROLL 224 + 224 = 448'.

TC-CLIN-PROPS-011 Bypass protection

Tests UR(s)	UR-CLIN-025
Command / action	`Rscript repos/clinical/R/jrc_clinical_ss_props.R ... (RENV_PATHS_ROOT unset)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_ss_survival

TC-CLIN-SURV-001 Superiority happy path — exit 0

Tests UR(s)	UR-CLIN-026
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6`
Rationale	Confirms the script accepts standard valid inputs and completes successfully.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'time-to-event endpoint'.

TC-CLIN-SURV-002 Superiority numeric correctness

Tests UR(s)	UR-CLIN-027
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6`
Rationale	Verifies the events formula: E = ceiling(4·(z0.975+z0.80)²/log(0.70)²) = 247; n/arm = ceiling((246.86/0.6)/2) = 206.
Pass criterion	Events required = 247, n treatment = 206, n control = 206, n TOTAL = 412.

TC-CLIN-SURV-003 Non-inferiority numeric correctness

Tests UR(s)	UR-CLIN-028
Command / action	`jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.0 --margin 1.3 --event-prob 0.5`
Rationale	Verifies the NI branch: E = ceiling(4·7.848886/log(1.3)²) = 457; n/arm = ceiling((456.10/0.5)/2) = 457.
Pass criterion	Events required = 457, n TOTAL = 914.

TC-CLIN-SURV-004 Equivalence not offered

Tests UR(s)	UR-CLIN-029
Command / action	`jrc_clinical_ss_survival --framework equivalence --power 0.80 --alpha 0.05 --sides 1 --hr 1.0 --event-prob 0.5`
Rationale	Equivalence is deliberately out of scope for time-to-event endpoints in this module.
Pass criterion	Exit code ≠ 0. 'not offered' in combined stdout/stderr.

TC-CLIN-SURV-005 Superiority with HR = 1 rejected

Tests UR(s)	UR-CLIN-030
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 1 --event-prob 0.6`
Rationale	HR = 1 means no effect; required events would be infinite.
Pass criterion	Exit code ≠ 0. 'different from 1' in combined stdout/stderr.

TC-CLIN-SURV-006 NI margin must exceed 1

Tests UR(s)	UR-CLIN-031
Command / action	`jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.0 --margin 0.9 --event-prob 0.5`
Rationale	An NI margin ≤ 1 is not a non-inferiority margin on a hazard ratio.
Pass criterion	Exit code ≠ 0. '--margin' in combined stdout/stderr.

TC-CLIN-SURV-007 NI needs true HR inside the margin

Tests UR(s)	UR-CLIN-032
Command / action	`jrc_clinical_ss_survival --framework non_inferiority --power 0.80 --alpha 0.025 --sides 1 --hr 1.4 --margin 1.3 --event-prob 0.5`
Rationale	A true HR at or beyond the margin cannot demonstrate non-inferiority at any n.
Pass criterion	Exit code ≠ 0. '--hr < --margin' in combined stdout/stderr.

TC-CLIN-SURV-008 --event-prob required

Tests UR(s)	UR-CLIN-033
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70`

Rationale	Events cannot be converted to subjects without the event probability.
Pass criterion	Exit code ≠ 0. '--event-prob' in combined stdout/stderr.

TC-CLIN-SURV-009 Sensitivity table over the event probability

Tests UR(s)	UR-CLIN-034
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6 --sensitivity`
Rationale	The event probability is the most uncertain design input; scenarios must be recomputed in the validated layer.
Pass criterion	5 scenario rows (0.48…0.72); assumed row marked.

TC-CLIN-SURV-010 Dropout inflation

Tests UR(s)	UR-CLIN-035
Command / action	`jrc_clinical_ss_survival --framework superiority --power 0.80 --alpha 0.05 --sides 2 --hr 0.70 --event-prob 0.6 --dropout 0.20`
Rationale	Enrolled N is a script responsibility: ceiling(206/0.80) = 258 per arm.
Pass criterion	Output contains 'ENROLL 258 + 258 = 516'.

TC-CLIN-SURV-011 Bypass protection

Tests UR(s)	UR-CLIN-036
Command / action	`Rscript repos/clinical/R/jrc_clinical_ss_survival.R ... (RENV_PATHS_ROOT unset)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code ≠ 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_dx_accuracy

TC-CLIN-DXACC-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-037
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv`
Rationale	Confirms the script accepts a standard valid dataset and completes.
Pass criterion	Exit code = 0. Output contains 'Diagnostic accuracy' and 'reference standard'.

TC-CLIN-DXACC-002 2x2 contingency table

Tests UR(s)	UR-CLIN-038
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv`
Rationale	The 2x2 is the basis of every downstream measure; an error here propagates to all of them.
Pass criterion	TP 79, FP 6 / FN 16, TN 99; margins 95 reference-positive, 105 reference-negative, 200 total.

TC-CLIN-DXACC-003 Sensitivity with exact interval

Tests UR(s)	UR-CLIN-039
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv`
Rationale	Sensitivity is the primary measure FDA (2007) requires with a CI.
Pass criterion	Sensitivity = 0.8316 (79/95), 95% CI (0.7410, 0.9006).

TC-CLIN-DXACC-004 Specificity with exact interval

Tests UR(s)	UR-CLIN-040
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv`
Rationale	Specificity is the co-primary measure FDA (2007) requires with a CI.
Pass criterion	Specificity = 0.9429 (99/105), 95% CI (0.8798, 0.9787).

TC-CLIN-DXACC-005 Overall accuracy with exact interval

Tests UR(s)	UR-CLIN-041
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv`
Rationale	Accuracy is reported for completeness; the report also carries the FDA caution against relying on it alone.
Pass criterion	Accuracy = 0.8900 (178/200), 95% CI (0.8382, 0.9298).

TC-CLIN-DXACC-006 Positive likelihood ratio with log interval

Tests UR(s)	UR-CLIN-042
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv`
Rationale	Likelihood ratios need an interval on the log scale; a Wald interval on the ratio itself can cover negative values.
Pass criterion	LR+ = 14.5526, 95% CI (6.6563, 31.8163).

TC-CLIN-DXACC-007 Negative likelihood ratio with log interval

Tests UR(s)	UR-CLIN-043
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv`
Rationale	As TC-CLIN-DXACC-006, for the negative likelihood ratio.
Pass criterion	LR- = 0.1786, 95% CI (0.1140, 0.2799).

TC-CLIN-DXACC-008 PPV/NPV at the study prevalence, with caveat

Tests UR(s)	UR-CLIN-044
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv`
Rationale	PPV/NPV read straight off a 2x2 are the most commonly misreported quantities in diagnostic submissions; the caveat must be unmissable.
Pass criterion	PPV = 0.9294 (79/85), NPV = 0.8609 (99/115); output states 'study prevalence (0.4750)' and 'valid ONLY if'.

TC-CLIN-DXACC-009 Wilson interval option

Tests UR(s)	UR-CLIN-045

Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --ci wilson`
Rationale	Wilson is the shorter interval with better mean coverage; it must sit strictly inside the exact interval it replaces.
Pass criterion	Sensitivity estimate unchanged at 0.8316; 95% CI (0.7438, 0.8936), strictly inside the exact (0.7410, 0.9006).

TC-CLIN-DXACC-010 Bayes-adjusted PPV/NPV at a stated prevalence

Tests UR(s)	UR-CLIN-046
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --prevalence 0.02`
Rationale	This is the correct reporting for a case-control or enriched design, per FDA (2007). It is also the clearest demonstration of why the unadjusted values mislead: the same test reads PPV 0.93 at the study's 47.5% prevalence and 0.23 at 2%.
Pass criterion	PPV (adj) = 0.2290, 95% CI (0.1117, 0.4637); NPV (adj) = 0.9964. The unadjusted PPV 0.9294 is still shown.

TC-CLIN-DXACC-011 Confidence level honoured

Tests UR(s)	UR-CLIN-047
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --conf 0.90`
Rationale	Confirms the confidence level reaches the interval maths rather than only the printed label.
Pass criterion	Point estimate unchanged; both 90% bounds strictly inside the corresponding 95% bounds.

TC-CLIN-DXACC-012 Non-standard labels via --positive

Tests UR(s)	UR-CLIN-048
Command / action	`jrc_clinical_dx_accuracy dx_custom_labels.csv --positive detected`
Rationale	Real assay exports rarely use 1/0; the escape hatch must work.
Pass criterion	Exit code = 0. Sensitivity = 0.8000 (4/5), specificity = 0.8000 (4/5).

TC-CLIN-DXACC-013 Rows with missing data dropped and reported

Tests UR(s)	UR-CLIN-049
Command / action	`jrc_clinical_dx_accuracy dx_missing_rows.csv`
Rationale	Silently dropping subjects would misstate the denominator; the count must be visible in the report.
Pass criterion	Exit code = 0. Output contains '4 evaluable' and '2 row(s) dropped'.

TC-CLIN-DXACC-014 Invalid --ci rejected

Tests UR(s)	UR-CLIN-050
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --ci bogus`
Rationale	Guards against a silent fallback to an unintended interval method.
Pass criterion	Exit code != 0. Output contains '--ci must be'.

TC-CLIN-DXACC-015 Out-of-range --prevalence rejected

Tests UR(s)	UR-CLIN-051
Command / action	`jrc_clinical_dx_accuracy dx_accuracy_n200_seed42.csv --prevalence 1.5`
Rationale	A prevalence outside (0,1) is meaningless and would silently produce nonsense Bayes values.
Pass criterion	Exit code != 0. Output names '--prevalence'.

TC-CLIN-DXACC-016 Missing required column rejected

Tests UR(s)	UR-CLIN-052
Command / action	`jrc_clinical_dx_accuracy dx_roc_n200_seed7.csv`
Rationale	The ROC fixture carries 'score' rather than 'result'; the error must name what is missing rather than fail obscurely.
Pass criterion	Exit code != 0. Output contains 'Missing column(s): result'.

TC-CLIN-DXACC-017 Duplicate subject id rejected

Tests UR(s)	UR-CLIN-053

Command / action	`jrc_clinical_dx_accuracy dx_dup_id.csv`
Rationale	A duplicated subject would be double-counted in the 2x2, inflating the apparent sample size.
Pass criterion	Exit code != 0. Output contains 'Duplicate id(s) found'.

TC-CLIN-DXACC-018 Unrecognised labels rejected

Tests UR(s)	UR-CLIN-054
Command / action	`jrc_clinical_dx_accuracy dx_bad_labels.csv`
Rationale	Guessing which of two unknown labels means 'positive' could silently invert sensitivity and specificity.
Pass criterion	Exit code != 0. Output contains 'does not recognise'.

TC-CLIN-DXACC-019 Absent reference group rejected

Tests UR(s)	UR-CLIN-055
Command / action	`jrc_clinical_dx_accuracy dx_all_reference_positive.csv`
Rationale	Specificity is undefined without reference-negative subjects; reporting 0/0 would be worse than refusing.
Pass criterion	Exit code != 0. Output contains 'specificity is undefined'.

TC-CLIN-DXACC-020 Bypass guardrail

Tests UR(s)	UR-CLIN-056
Command / action	`Rscript --vanilla jrc_clinical_dx_accuracy.R (no RENV_PATHS_ROOT)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code != 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_dx_ss

TC-CLIN-DXSS-001 Precision method happy path — exit 0

Tests UR(s)	UR-CLIN-057
Command / action	`jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10`
Rationale	Confirms the script accepts standard valid inputs and completes.
Pass criterion	Exit code = 0. Output contains 'Clinical sample size' and 'diagnostic accuracy'.

TC-CLIN-DXSS-002 Precision numeric correctness (Buderer 1996)

Tests UR(s)	UR-CLIN-058
Command / action	`jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10`
Rationale	Reproduces the published worked example: n_pos = z^2*0.9*0.1/0.05^2 = 138.29 -> 139; n_neg = z^2*0.95*0.05/0.05^2 = 72.99 -> 73; N = 138.29/0.10 = 1382.93 -> 1383.
Pass criterion	n reference + = 139, n reference - = 73, N TOTAL = 1383.

TC-CLIN-DXSS-003 Hypothesis numeric correctness

Tests UR(s)	UR-CLIN-059
Command / action	`jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --sens-goal 0.80 --spec-goal 0.90 --power 0.80 --alpha 0.025 --sides 1 --prevalence 0.10`
Rationale	n_pos = (z_a*sqrt(.8*.2) + z_b*sqrt(.9*.1))^2/(.9-.8)^2 = 107.43 -> 108; n_neg = (z_a*sqrt(.9*.1) + z_b*sqrt(.95*.05))^2/(.95-.90)^2 = 238.02 -> 239; N = max(107.43/.10, 238.02/.90) -> 1075.
Pass criterion	n reference + = 108, n reference - = 239, N TOTAL = 1075.

TC-CLIN-DXSS-004 Binding arm identified

Tests UR(s)	UR-CLIN-060
Command / action	`jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10`
Rationale	At low prevalence the reference-positive arm binds; naming it tells the designer which assumption to attack to shrink the study.
Pass criterion	Output contains 'Binding arm : sensitivity (reference-positive)'.

TC-CLIN-DXSS-005 Rarer prevalence raises enrolment, not arm size

Tests UR(s)	UR-CLIN-061
Command / action	`jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.05`
Rationale	Separates the two effects: arm requirements depend only on the target precision; only the enrolment needed to reach them depends on prevalence. Halving prevalence doubles enrolment.
Pass criterion	n reference + = 139 (unchanged); N TOTAL = 2766, greater than the 1383 of TC-CLIN-DXSS-002.

TC-CLIN-DXSS-006 Prevalence sensitivity table

Tests UR(s)	UR-CLIN-062
Command / action	`jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10 --sensitivity`
Rationale	Prevalence is the assumption most often wrong and most punishing when it is; the table makes the exposure visible at design time.
Pass criterion	Exit code = 0. Five rows (prevalence x 0.5, 0.75, 1.0, 1.5, 2.0); the assumed row marked '<- assumed'.

TC-CLIN-DXSS-007 Dropout inflation

Tests UR(s)	UR-CLIN-063
Command / action	`jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --sens-goal 0.80 --spec-goal 0.90 --power 0.80 --alpha 0.025 --sides 1 --prevalence 0.10 --dropout 0.10`
Rationale	enrolled = ceiling(1075 / 0.90) = 1195.

Pass criterion	Output contains 'ENROLL 1195 subjects'.

TC-CLIN-DXSS-008 Sidedness default differs by method

Tests UR(s)	UR-CLIN-064
Command / action	`jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 0.10  / jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --sens-goal 0.80 --spec-goal 0.90 --power 0.80 --prevalence 0.10`
Rationale	A CI half-width is inherently two-sided; a performance-goal test is conventionally one-sided. Defaulting wrongly would silently mis-size the study.
Pass criterion	Precision output contains 'two-sided, z = 1.9600'; hypothesis output without explicit --sides contains '1-sided'.

TC-CLIN-DXSS-009 Precision requires --halfwidth

Tests UR(s)	UR-CLIN-065
Command / action	`jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --prevalence 0.10`
Rationale	There is no defensible default target precision; it must be stated.
Pass criterion	Exit code != 0. Output contains 'requires --halfwidth'.

TC-CLIN-DXSS-010 Hypothesis requires --sens-goal

Tests UR(s)	UR-CLIN-066
Command / action	`jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --spec-goal 0.90 --prevalence 0.10`
Rationale	There is no defensible default performance goal; it must be stated.
Pass criterion	Exit code != 0. Output contains 'requires --sens-goal'.

TC-CLIN-DXSS-011 Goal must be below the expected value

Tests UR(s)	UR-CLIN-067
Command / action	`jrc_clinical_dx_ss --method hypothesis --sens-expected 0.90 --spec-expected 0.95 --sens-goal 0.95 --spec-goal 0.90 --prevalence 0.10`
Rationale	A goal at or above the expected value cannot be powered against; the formula would divide by a non-positive difference.
Pass criterion	Exit code != 0. Output contains '--sens-goal must be strictly less than'.

TC-CLIN-DXSS-012 Out-of-range --prevalence rejected

Tests UR(s)	UR-CLIN-068
Command / action	`jrc_clinical_dx_ss --method precision --sens-expected 0.90 --spec-expected 0.95 --halfwidth 0.05 --prevalence 1.2`
Rationale	A prevalence outside (0,1) is meaningless and would produce a negative or infinite enrolment.
Pass criterion	Exit code != 0. Output names '--prevalence'.

TC-CLIN-DXSS-013 Invalid --method rejected

Tests UR(s)	UR-CLIN-069
Command / action	`jrc_clinical_dx_ss --method bogus`
Rationale	Guards against a silent fallback to an unintended sizing method.
Pass criterion	Exit code != 0. Output contains '--method must be one of'.

TC-CLIN-DXSS-014 Bypass guardrail

Tests UR(s)	UR-CLIN-070
Command / action	`Rscript --vanilla jrc_clinical_dx_ss.R (no RENV_PATHS_ROOT)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code != 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_dx_roc

TC-CLIN-DXROC-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-071
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv`
Rationale	Confirms the script accepts a standard valid dataset and completes.
Pass criterion	Exit code = 0. Output contains 'ROC analysis'; 80 reference-positive and 120 reference-negative subjects reported.

TC-CLIN-DXROC-002 ROC plot written to ~/Downloads/

Tests UR(s)	UR-CLIN-072
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv`
Rationale	The plot is a deliverable of the script, not a side effect.
Pass criterion	A file matching *_jrc_clinical_dx_roc.png is created in ~/Downloads/ during the run.

TC-CLIN-DXROC-003 PUBLISHED ANCHOR — Hanley & McNeil (1982) AUC

Tests UR(s)	UR-CLIN-073
Command / action	`jrc_clinical_dx_roc dx_roc_hanley_mcneil_1982.csv`
Rationale	The AUC and its DeLong variance are implemented in base R in this project rather than delegated to a package, so the acceptance value is taken from the ORIGINAL LITERATURE rather than from our own output. Hanley JA, McNeil BJ (1982), Radiology 143:29-36, Table 1: 51 abnormal and 58 normal cases on a 5-point diagnostic confidence rating; the paper reports an area of 0.893. Note the paper's SE is deliberately NOT used as an acceptance criterion: it is a binormal-based estimator, whereas this script reports the nonparametric DeLong SE.
Pass criterion	51 reference-positive, 58 reference-negative. AUC = 0.8932, which equals the published 0.893 at the paper's three decimal places.

TC-CLIN-DXROC-004 AUC, DeLong SE and confidence interval

Tests UR(s)	UR-CLIN-074
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv`
Rationale	The DeLong interval is the conventional nonparametric interval for an empirical AUC.
Pass criterion	AUC = 0.8268, SE = 0.0294, 95% CI (0.7691, 0.8845), labelled 'DeLong (raw scale)'.

TC-CLIN-DXROC-005 Tie-aware AUC kernel

Tests UR(s)	UR-CLIN-075
Command / action	`jrc_clinical_dx_roc dx_roc_ties_n26_seed11.csv`
Rationale	Integer scores 1-5 across 26 subjects make almost every comparison a tie. The expected value is obtainable only with the psi = 0.5 kernel; a kernel that ignored ties would not return it. This is what makes the empirical AUC equal the trapezoidal area under the ROC curve.
Pass criterion	12 reference-positive, 14 reference-negative. AUC = 0.5952.

TC-CLIN-DXROC-006 Youden-optimal operating point

Tests UR(s)	UR-CLIN-076
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv`
Rationale	Youden's J is the conventional summary of the best single cutoff.
Pass criterion	J = 0.5125, sensitivity = 0.5875, specificity = 0.9250.

TC-CLIN-DXROC-007 Cutoff reported as an observed value with a rule

Tests UR(s)	UR-CLIN-077
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv`
Rationale	An observed value is directly reportable and cannot fall outside the measured range. Other tools print the midpoint between adjacent observed values; both name the identical classifier, but only an explicit rule makes the report unambiguous.
Pass criterion	Output contains 'Cutoff : score >= 2.192 => positive'.

TC-CLIN-DXROC-008 Test of AUC = 0.5

Tests UR(s)	UR-CLIN-078
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv`
Rationale	AUC = 0.5 is the no-discrimination null; the test is the basic question asked of any new marker.
Pass criterion	Output contains 'H0: AUC = 0.5' and 'z = 11.1017'.

TC-CLIN-DXROC-009 Direction reverses the AUC about 0.5

Tests UR(s)	UR-CLIN-079
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv --direction lower`
Rationale	Reversing the assumed assay orientation must map AUC to 1 - AUC exactly; any other result indicates the orientation is not applied consistently.
Pass criterion	AUC = 0.1732 against 0.8268 for the default direction; the two sum to 1 within 1e-9.

TC-CLIN-DXROC-010 Degenerate cutoff reported honestly

Tests UR(s)	UR-CLIN-080
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv --direction lower`
Rationale	With the direction reversed, every candidate cutoff has J <= 0, so the Youden optimum falls on an infinite sentinel threshold. Printing 'score <= Inf => positive' would be meaningless; the honest report is that there is no usable cutoff, plus a warning that the assay orientation is probably inverted.
Pass criterion	Exit code = 0. Output contains 'Cutoff : none' and 'AUC < 0.5', and suggests '--direction higher'. No infinite threshold is printed in the Youden section.

TC-CLIN-DXROC-011 Logit interval option

Tests UR(s)	UR-CLIN-081
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv --ci-method logit`
Rationale	The logit interval stays inside (0,1) and behaves better when the AUC approaches 1, where the raw-scale interval can overshoot.
Pass criterion	AUC unchanged at 0.8268; 95% CI (0.7614, 0.8772), labelled 'DeLong (logit-transformed)'.

TC-CLIN-DXROC-012 Confidence level honoured

Tests UR(s)	UR-CLIN-082
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv --conf 0.90`
Rationale	Confirms the confidence level reaches the interval maths rather than only the printed label.
Pass criterion	AUC unchanged at 0.8268; 90% CI (0.7784, 0.8752), strictly inside the 95% CI (0.7691, 0.8845).

TC-CLIN-DXROC-013 Invalid --direction rejected

Tests UR(s)	UR-CLIN-083
Command / action	`jrc_clinical_dx_roc dx_roc_n200_seed7.csv --direction sideways`
Rationale	Guards against a silent fallback to an unintended orientation, which would invert the entire analysis.
Pass criterion	Exit code != 0. Output contains '--direction must be'.

TC-CLIN-DXROC-014 Missing required column rejected

Tests UR(s)	UR-CLIN-084
Command / action	`jrc_clinical_dx_roc dx_accuracy_n200_seed42.csv`
Rationale	The accuracy fixture carries 'result' rather than 'score'; the error must name what is missing rather than fail obscurely.
Pass criterion	Exit code != 0. Output contains 'Missing column(s): score'.

TC-CLIN-DXROC-015 Duplicate subject id rejected

Tests UR(s)	UR-CLIN-085
Command / action	`jrc_clinical_dx_roc dx_dup_id.csv`
Rationale	A duplicated subject would be counted twice in the rank statistics, biasing the AUC and understating its variance.
Pass criterion	Exit code != 0. Output contains 'Duplicate id(s) found'.

TC-CLIN-DXROC-016 Absent reference group rejected

Tests UR(s)	UR-CLIN-086
Command / action	`jrc_clinical_dx_roc dx_all_reference_positive.csv`
Rationale	An ROC curve is undefined without both reference groups; there is nothing to discriminate between.
Pass criterion	Exit code != 0. Output contains 'At least 2 reference-negative'.

TC-CLIN-DXROC-017 Bypass guardrail

Tests UR(s)	UR-CLIN-087
Command / action	`Rscript --vanilla jrc_clinical_dx_roc.R (no RENV_PATHS_ROOT)`
Rationale	Confirms that running the R script outside the validated jrrun path stops with the guardrail message.
Pass criterion	Exit code != 0. 'RENV_PATHS_ROOT' in combined stdout/stderr.

Test Cases — jrc_clinical_km

TC-CLIN-KM-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-088
Command / action	`jrc_clinical_km km_aml_miller1981.csv --group group`
Rationale	Confirms the script accepts a standard right-censored dataset and completes.
Pass criterion	Exit code = 0. Output contains 'Kaplan-Meier survival analysis'; 23 subjects reported.

TC-CLIN-KM-002 Survival plot written to ~/Downloads/

Tests UR(s)	UR-CLIN-089
Command / action	`jrc_clinical_km km_aml_miller1981.csv --group group`
Rationale	The plot is a deliverable of the script, not a side effect.
Pass criterion	A file matching *_jrc_clinical_km.png is created during the run.

TC-CLIN-KM-003 PUBLISHED ANCHOR — Miller (1981) median survival

Tests UR(s)	UR-CLIN-090
Command / action	`jrc_clinical_km km_aml_miller1981.csv --group group`
Rationale	The Kaplan-Meier estimate is anchored to the ORIGINAL LITERATURE, not to our own output. Miller RG (1981), Survival Analysis, uses the aml data (11 maintained, 12 non-maintained patients); the maintained median is 31 weeks and the non-maintained 23 weeks.
Pass criterion	Maintained arm median = 31; non-maintained median = 23.

TC-CLIN-KM-004 PUBLISHED ANCHOR — Miller (1981) log-rank test

Tests UR(s)	UR-CLIN-091
Command / action	`jrc_clinical_km km_aml_miller1981.csv --group group`
Rationale	The log-rank test is anchored to the published analysis: chi-square 3.4 on 1 degree of freedom, p = 0.07.
Pass criterion	Chi-square = 3.3964 (rounds to the published 3.4), df = 1, p = 0.065 (rounds to 0.07).

TC-CLIN-KM-005 Log-rank observed vs expected events

Tests UR(s)	UR-CLIN-092
Command / action	`jrc_clinical_km km_aml_miller1981.csv --group group`
Rationale	Observed/expected are the components of the log-rank statistic and make the direction of the effect auditable.
Pass criterion	Maintained observed = 7, expected = 10.69; non-maintained observed = 11, expected = 7.31.

TC-CLIN-KM-006 Ungrouped overall median and CI

Tests UR(s)	UR-CLIN-093
Command / action	`jrc_clinical_km km_aml_miller1981.csv`
Rationale	The single-cohort path must produce a valid summary.
Pass criterion	Overall n = 23; median survival 27 with 95% CI (18, 45).

TC-CLIN-KM-007 Survival probability at a time point

Tests UR(s)	UR-CLIN-094
Command / action	`jrc_clinical_km km_aml_miller1981.csv --group group --time-point 30`
Rationale	S(t) is a primary reporting quantity for a survival study.
Pass criterion	S(30) = 0.6136 (maintained) and 0.2917 (non-maintained).

TC-CLIN-KM-008 Confidence level narrows the median CI

Tests UR(s)	UR-CLIN-095
Command / action	`jrc_clinical_km km_aml_miller1981.csv --conf 0.90`
Rationale	A lower confidence level must produce a strictly tighter interval.
Pass criterion	90% median CI = (18, 43), inside the 95% CI (18, 45).

TC-CLIN-KM-009 Rho-weighted (Peto) test differs from log-rank

Tests UR(s)	UR-CLIN-096
Command / action	`jrc_clinical_km km_aml_miller1981.csv --group group --rho 1`
Rationale	Confirms the weighting argument reaches survdiff and changes the result.
Pass criterion	Output names the Peto weighting; chi-square = 2.7793, distinct from the log-rank 3.3964.

TC-CLIN-KM-010 Event-label override

Tests UR(s)	UR-CLIN-097
Command / action	`jrc_clinical_km km_yesno.csv --event-positive yes`
Rationale	Allows non-standard event coding to be analysed without editing the file.
Pass criterion	Exit 0; 6 subjects reported.

TC-CLIN-KM-011 Yes/no event labels auto-detected

Tests UR(s)	UR-CLIN-098
Command / action	`jrc_clinical_km km_yesno.csv`
Rationale	Confirms the label auto-detection covers yes/no as well as 1/0.
Pass criterion	Exit 0; 3 events reported.

TC-CLIN-KM-012 Median CI reported as NA when not reached

Tests UR(s)	UR-CLIN-099
Command / action	`jrc_clinical_km km_aml_miller1981.csv --group group`
Rationale	An unreached bound is a real result and must be shown as NA, not fabricated.
Pass criterion	The maintained-arm median CI upper bound is reported as NA.

TC-CLIN-KM-013 Missing required column rejected

Tests UR(s)	UR-CLIN-100
Command / action	`jrc_clinical_km dx_accuracy_n200_seed42.csv`
Rationale	Fail-closed on a malformed input.
Pass criterion	Non-zero exit; message contains 'Missing column(s)'.

TC-CLIN-KM-014 Duplicate id rejected

Tests UR(s)	UR-CLIN-101
Command / action	`jrc_clinical_km km_dup_id.csv`
Rationale	Each subject must appear exactly once.
Pass criterion	Non-zero exit; message contains 'Duplicate id(s) found'.

TC-CLIN-KM-015 All-censored data rejected

Tests UR(s)	UR-CLIN-102
Command / action	`jrc_clinical_km km_all_censored.csv`
Rationale	Survival cannot be estimated with zero events.
Pass criterion	Non-zero exit; message contains 'No events observed'.

TC-CLIN-KM-016 Single-group comparison rejected

Tests UR(s)	UR-CLIN-103
Command / action	`jrc_clinical_km km_one_group.csv --group arm`
Rationale	A comparison needs at least two groups.
Pass criterion	Non-zero exit; message contains 'only one distinct value'.

TC-CLIN-KM-017 Unrecognised event labels rejected

Tests UR(s)	UR-CLIN-104
Command / action	`jrc_clinical_km km_bad_event.csv`
Rationale	Ambiguous event coding must fail rather than be guessed.
Pass criterion	Non-zero exit; message contains 'does not recognise'.

TC-CLIN-KM-018 Non-positive time rejected

Tests UR(s)	UR-CLIN-105
Command / action	`jrc_clinical_km km_neg_time.csv`

Rationale	Survival time must be strictly positive.
Pass criterion	Non-zero exit; message contains 'strictly positive'.

TC-CLIN-KM-019 Direct execution without RENV_PATHS_ROOT blocked

Tests UR(s)	UR-CLIN-106
Command / action	`Rscript repos/clinical/R/jrc_clinical_km.R (no RENV_PATHS_ROOT)`
Rationale	The environment guard must stop accidental un-validated execution.
Pass criterion	Non-zero exit; message contains 'RENV_PATHS_ROOT'.

Test Cases — jrc_clinical_coxph

TC-CLIN-COX-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-107
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex`
Rationale	Confirms the script fits a standard Cox model and completes.
Pass criterion	Exit code = 0. Output contains 'Cox proportional-hazards regression'; 228 subjects, 165 events.

TC-CLIN-COX-002 Schoenfeld plot written to ~/Downloads/

Tests UR(s)	UR-CLIN-108
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex`
Rationale	The diagnostic plot is a deliverable of the script.
Pass criterion	A file matching *_jrc_clinical_coxph.png is created during the run.

TC-CLIN-COX-003 PUBLISHED ANCHOR — Therneau & Grambsch sex HR

Tests UR(s)	UR-CLIN-109
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex`
Rationale	The Cox fit is anchored to the ORIGINAL LITERATURE, not to our own output. Therneau & Grambsch (2000), Modeling Survival Data, analyse the lung data with Surv(time,status) ~ age + sex (sex 1 = male, 2 = female) and report a female-vs-male hazard ratio near 0.60.
Pass criterion	sex HR = 0.5986 with 95% CI (0.4311, 0.8311).

TC-CLIN-COX-004 PUBLISHED ANCHOR — Therneau & Grambsch age HR

Tests UR(s)	UR-CLIN-110
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex`
Rationale	Second published coefficient from the same reference analysis.
Pass criterion	age HR = 1.0172 with 95% CI (0.9990, 1.0357).

TC-CLIN-COX-005 Harrell's concordance

Tests UR(s)	UR-CLIN-111
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex`
Rationale	Concordance is the standard discrimination measure for a Cox model.
Pass criterion	Concordance C = 0.6029.

TC-CLIN-COX-006 Global likelihood-ratio and Wald tests

Tests UR(s)	UR-CLIN-112
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex`
Rationale	The global tests establish overall model significance.
Pass criterion	Likelihood-ratio chi-square = 14.1231; Wald chi-square = 13.4700.

TC-CLIN-COX-007 PH assumption holds — reported as OK

Tests UR(s)	UR-CLIN-113
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex`
Rationale	The PH test governs whether a single HR is interpretable; a passing case must be reported as such.
Pass criterion	Output contains 'PH assumption OK'; global Schoenfeld p = 0.2502.

TC-CLIN-COX-008 PUBLISHED ANCHOR — PH violation flagged (veteran/karno)

Tests UR(s)	UR-CLIN-114
Command / action	`jrc_clinical_coxph coxph_veteran_karno.csv --covariates karno`
Rationale	The veteran data with the Karnofsky score is the textbook PH violation (Therneau & Grambsch). The script must detect and flag it, not report a misleading constant HR.
Pass criterion	Output contains 'PH assumption VIOLATED'; global Schoenfeld p = 0.000316.

TC-CLIN-COX-009 Factor (categorical) covariate

Tests UR(s)	UR-CLIN-115
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates sex_label`

Rationale	Categorical covariates must produce per-level hazard ratios; the male-vs-female HR is the reciprocal of the numeric-sex HR.
Pass criterion	sex_labelmale HR = 1.7007 (= 1 / 0.5986).

TC-CLIN-COX-010 Rows with missing covariate values dropped

Tests UR(s)	UR-CLIN-116
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,ph_ecog`
Rationale	A complete-case fit is standard for Cox regression; the drop count must be reported.
Pass criterion	228 - 1 = 227 subjects fitted; output reports 'Dropped rows : 1'.

TC-CLIN-COX-011 Breslow ties differ from Efron

Tests UR(s)	UR-CLIN-117
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex --ties breslow`
Rationale	Confirms the tie argument reaches coxph and changes the estimate.
Pass criterion	sex HR = 0.5990 under breslow, distinct from the efron 0.5986.

TC-CLIN-COX-012 Confidence level narrows the HR CI

Tests UR(s)	UR-CLIN-118
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates age,sex --conf 0.90`
Rationale	A lower confidence level must produce a strictly tighter interval.
Pass criterion	sex HR CI = (0.4545, 0.7884) at 90%, inside the 95% (0.4311, 0.8311).

TC-CLIN-COX-013 Missing covariate column rejected

Tests UR(s)	UR-CLIN-119
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv --covariates nosuchcol`
Rationale	Fail-closed on a covariate name the file does not contain.
Pass criterion	Non-zero exit; message contains 'Missing column(s)'.

TC-CLIN-COX-014 --covariates required

Tests UR(s)	UR-CLIN-120
Command / action	`jrc_clinical_coxph coxph_lung_therneau.csv`
Rationale	A Cox model needs at least one covariate; omission must fail clearly.
Pass criterion	Non-zero exit; message contains '--covariates is required'.

TC-CLIN-COX-015 Duplicate id rejected

Tests UR(s)	UR-CLIN-121
Command / action	`jrc_clinical_coxph km_dup_id.csv --covariates time`
Rationale	Each subject must appear exactly once.
Pass criterion	Non-zero exit; message contains 'Duplicate id(s) found'.

TC-CLIN-COX-016 Fewer than two events rejected

Tests UR(s)	UR-CLIN-122
Command / action	`jrc_clinical_coxph coxph_few_events.csv --covariates age`
Rationale	A Cox model cannot be fit reliably with too few events.
Pass criterion	Non-zero exit; message contains 'Fewer than 2 events'.

TC-CLIN-COX-017 Direct execution without RENV_PATHS_ROOT blocked

Tests UR(s)	UR-CLIN-123
Command / action	`Rscript repos/clinical/R/jrc_clinical_coxph.R (no RENV_PATHS_ROOT)`
Rationale	The environment guard must stop accidental un-validated execution.
Pass criterion	Non-zero exit; message contains 'RENV_PATHS_ROOT'.

Test Cases — jrc_clinical_dx_compare

TC-CLIN-DXCMP-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-124
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv`
Rationale	Confirms the script accepts a standard paired dataset and completes.
Pass criterion	Exit 0; output contains 'Paired ROC comparison'; 70 reference-positive and 100 reference-negative reported.

TC-CLIN-DXCMP-002 Comparison plot written to ~/Downloads/

Tests UR(s)	UR-CLIN-125
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv`
Rationale	The plot is a deliverable of the script.
Pass criterion	A file matching *_jrc_clinical_dx_compare.png is created during the run.

TC-CLIN-DXCMP-003 EXTERNAL-ORACLE ANCHOR — per-test AUCs

Tests UR(s)	UR-CLIN-126
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv`
Rationale	The two AUCs, their covariance and the paired statistic are implemented in base R here (DeLong, DeLong & Clarke-Pearson 1988, Biometrics 44:837-845 — the method written for the correlated-curve case, see also Hanley & McNeil 1983). The acceptance values are therefore taken from an INDEPENDENT reference implementation, pROC 1.19.0.1 roc.test(method='delong'), not from our own output. pROC is not a project dependency and is not pinned; it was used once as an oracle, as for jrc_clinical_dx_roc.
Pass criterion	AUC A (test_a) = 0.8681 (95% CI 0.8138, 0.9225); AUC B (test_b) = 0.6670 (95% CI 0.5851, 0.7489).

TC-CLIN-DXCMP-004 Difference in AUCs with CI

Tests UR(s)	UR-CLIN-127
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv`
Rationale	The difference and its CI are the primary output of a paired comparison.
Pass criterion	Difference = +0.2011 with 95% CI (+0.1148, +0.2875).

TC-CLIN-DXCMP-005 Paired DeLong test statistic

Tests UR(s)	UR-CLIN-128
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv`
Rationale	The correlation-corrected test is the whole point of the script; it must match the reference implementation.
Pass criterion	z = 4.5640, p = 5.019e-06 (bit-identical to pROC roc.test: z to ~3e-15, p to ~6e-20).

TC-CLIN-DXCMP-006 Between-AUC correlation reported

Tests UR(s)	UR-CLIN-129
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv`
Rationale	The correlation is what distinguishes the paired test from two independent analyses; reporting it makes the correction visible.
Pass criterion	Correlation = 0.2468.

TC-CLIN-DXCMP-007 Significance statement

Tests UR(s)	UR-CLIN-130
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv`
Rationale	A plain-language conclusion prevents misreading the numbers.
Pass criterion	Output states test_a has the higher AUC and the difference is significant at the 95% level.

TC-CLIN-DXCMP-008 Confidence level narrows the difference CI

Tests UR(s)	UR-CLIN-131
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv --conf 0.90`
Rationale	A lower confidence level must give a strictly tighter interval.

Pass criterion	90% difference CI = (+0.1287, +0.2736), inside the 95% CI.

TC-CLIN-DXCMP-009 Logit per-test CI differs, stays in (0,1)

Tests UR(s)	UR-CLIN-132
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv --ci-method logit`
Rationale	Confirms the CI-method flag reaches the per-test interval and differs from the raw scale.
Pass criterion	AUC A estimate unchanged 0.8681; logit CI (0.8037, 0.9137), differing from the raw (0.8138, 0.9225); output names 'logit-transformed'.

TC-CLIN-DXCMP-010 Direction lower reflects both AUCs, flips z

Tests UR(s)	UR-CLIN-133
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv --direction lower`
Rationale	The direction is a claim about both assays; reversing it reflects both AUCs and the sign of the difference.
Pass criterion	AUC A = 0.1319, AUC B = 0.3330; z = -4.5640 (|z| unchanged).

TC-CLIN-DXCMP-011 Explicit --tests names the columns

Tests UR(s)	UR-CLIN-134
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv --tests test_a,test_b`
Rationale	Explicit column selection must work when the file has extra columns.
Pass criterion	Exit 0; output reports 'A = test_a vs B = test_b'.

TC-CLIN-DXCMP-012 Two test columns auto-inferred

Tests UR(s)	UR-CLIN-135
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv`
Rationale	Reduces friction for the common two-column case.
Pass criterion	Without --tests, test_a and test_b are used.

TC-CLIN-DXCMP-013 Ambiguous columns rejected

Tests UR(s)	UR-CLIN-136
Command / action	`jrc_clinical_dx_compare dx_compare_three_tests.csv`
Rationale	The script must not guess which two of three columns to compare.
Pass criterion	Non-zero exit; message contains 'Could not infer the two test columns'.

TC-CLIN-DXCMP-014 Named column absent rejected

Tests UR(s)	UR-CLIN-137
Command / action	`jrc_clinical_dx_compare dx_compare_paired_seed7.csv --tests test_a,nosuchcol`
Rationale	Fail-closed on a mistyped column name.
Pass criterion	Non-zero exit; message contains 'Test column(s) not found: nosuchcol'.

TC-CLIN-DXCMP-015 Missing reference column rejected

Tests UR(s)	UR-CLIN-138
Command / action	`jrc_clinical_dx_compare km_aml_miller1981.csv`
Rationale	Fail-closed on a malformed input.
Pass criterion	Non-zero exit; message contains 'Missing column(s): reference'.

TC-CLIN-DXCMP-016 Duplicate id rejected

Tests UR(s)	UR-CLIN-139
Command / action	`jrc_clinical_dx_compare dx_compare_dup_id.csv`
Rationale	Paired data requires each subject once.
Pass criterion	Non-zero exit; message contains 'Duplicate id(s) found'.

TC-CLIN-DXCMP-017 No reference-negative subjects rejected

Tests UR(s)	UR-CLIN-140
Command / action	`jrc_clinical_dx_compare dx_compare_all_positive.csv`
Rationale	An ROC curve is undefined without negatives.
Pass criterion	Non-zero exit; message contains 'At least 2 reference-negative'.

TC-CLIN-DXCMP-018 Direct execution without RENV_PATHS_ROOT blocked

Tests UR(s)	UR-CLIN-141
Command / action	`Rscript repos/clinical/R/jrc_clinical_dx_compare.R (no RENV_PATHS_ROOT)`
Rationale	The environment guard must stop accidental un-validated execution.
Pass criterion	Non-zero exit; message contains 'RENV_PATHS_ROOT'.

Test Cases — jrc_clinical_compare_means

TC-CLIN-CMEAN-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-142
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv`
Rationale	Confirms the script accepts a standard dataset and completes.
Pass criterion	Exit 0; 'Comparison of a continuous outcome between groups'; 20 subjects.

TC-CLIN-CMEAN-002 PUBLISHED ANCHOR — Student (1908) Welch t-test

Tests UR(s)	UR-CLIN-143
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv`
Rationale	The two-group comparison is anchored to the ORIGINAL dataset on which the t-test was devised (shipped with R as 'sleep'). Base R computes the statistic; the acceptance value is the documented result.
Pass criterion	t = -1.8608, df = 17.7765, p = 0.0794.

TC-CLIN-CMEAN-003 Mean difference and confidence interval

Tests UR(s)	UR-CLIN-144
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv`
Rationale	The effect and its precision are the primary output.
Pass criterion	Mean difference -1.5800 with 95% CI (-3.3655, +0.2055).

TC-CLIN-CMEAN-004 Standardised effect size

Tests UR(s)	UR-CLIN-145
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv`
Rationale	Effect size complements the p-value.
Pass criterion	Cohen's d = -0.8322, Hedges' g = -0.7970.

TC-CLIN-CMEAN-005 Mann-Whitney nonparametric alternative

Tests UR(s)	UR-CLIN-146
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv`
Rationale	A distribution-free alternative to the t-test.
Pass criterion	W = 25.5, p = 0.06582.

TC-CLIN-CMEAN-006 Student option (equal variance)

Tests UR(s)	UR-CLIN-147
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv --test student`
Rationale	Confirms the flag reaches t.test and changes the degrees of freedom.
Pass criterion	Student's t-test; t = -1.8608, df = 18, p = 0.07919.

TC-CLIN-CMEAN-007 Confidence level narrows the difference CI

Tests UR(s)	UR-CLIN-148
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv --conf 0.90`
Rationale	A lower confidence level must give a strictly tighter interval.
Pass criterion	90% CI (-3.0534, -0.1066), inside the 95% CI.

TC-CLIN-CMEAN-008 Per-group normality note

Tests UR(s)	UR-CLIN-149
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv`
Rationale	Guides the reader on t-test vs rank-test choice.
Pass criterion	Output contains 'Shapiro p =' for each group.

TC-CLIN-CMEAN-009 PUBLISHED ANCHOR — PlantGrowth one-way ANOVA

Tests UR(s)	UR-CLIN-150
Command / action	`jrc_clinical_compare_means means_plantgrowth.csv`

Rationale	The three-group case is anchored to the PlantGrowth data (shipped with R).
Pass criterion	F = 4.8461, df = 2, 27, p = 0.0159.

TC-CLIN-CMEAN-010 Missing required column rejected

Tests UR(s)	UR-CLIN-151
Command / action	`jrc_clinical_compare_means km_aml_miller1981.csv`
Rationale	Fail-closed on malformed input.
Pass criterion	Non-zero exit; 'Missing column(s)'.

TC-CLIN-CMEAN-011 Duplicate id rejected

Tests UR(s)	UR-CLIN-152
Command / action	`jrc_clinical_compare_means cmp_dup_id.csv`
Rationale	Each subject appears once.
Pass criterion	Non-zero exit; 'Duplicate id(s) found'.

TC-CLIN-CMEAN-012 Single group rejected

Tests UR(s)	UR-CLIN-153
Command / action	`jrc_clinical_compare_means cmp_one_group.csv`
Rationale	A comparison needs two groups.
Pass criterion	Non-zero exit; 'At least two groups'.

TC-CLIN-CMEAN-013 Group with too few observations rejected

Tests UR(s)	UR-CLIN-154
Command / action	`jrc_clinical_compare_means cmp_one_per_group.csv`
Rationale	A variance needs at least two points.
Pass criterion	Non-zero exit; 'fewer than 2 observations'.

TC-CLIN-CMEAN-014 Bad --test value rejected

Tests UR(s)	UR-CLIN-155
Command / action	`jrc_clinical_compare_means means_sleep_student1908.csv --test bogus`
Rationale	Fail-closed on an invalid option.
Pass criterion	Non-zero exit; '--test must be'.

TC-CLIN-CMEAN-015 Direct execution without RENV_PATHS_ROOT blocked

Tests UR(s)	UR-CLIN-156
Command / action	`Rscript repos/clinical/R/jrc_clinical_compare_means.R (no RENV_PATHS_ROOT)`
Rationale	The environment guard stops un-validated execution.
Pass criterion	Non-zero exit; 'RENV_PATHS_ROOT'.

Test Cases — jrc_clinical_compare_props

TC-CLIN-CPROP-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-157
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv`
Rationale	Confirms the script accepts a standard dataset and completes.
Pass criterion	Exit 0; 'Comparison of a categorical outcome'; contingency table shown.

TC-CLIN-CPROP-002 CLOSED-FORM ANCHOR — chi-square (Yates)

Tests UR(s)	UR-CLIN-158
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv`
Rationale	The 2x2 (treated 15/85, control 30/70) has hand-verifiable statistics.
Pass criterion	X2 = 5.6201, df = 1, p = 0.01776.

TC-CLIN-CPROP-003 Fisher's exact test

Tests UR(s)	UR-CLIN-159
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv`
Rationale	The exact alternative to the chi-square approximation.
Pass criterion	Fisher p = 0.01715.

TC-CLIN-CPROP-004 CLOSED-FORM ANCHOR — risk difference

Tests UR(s)	UR-CLIN-160
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv`
Rationale	RD = 15/100 - 30/100 = -0.15, a hand-derived value; Wald interval.
Pass criterion	Risk difference -0.1500 (95% CI -0.2639, -0.0361).

TC-CLIN-CPROP-005 CLOSED-FORM ANCHOR — risk ratio

Tests UR(s)	UR-CLIN-161
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv`
Rationale	RR = 0.15/0.30 = 0.50; Woolf-log interval.
Pass criterion	Risk ratio 0.5000 (95% CI 0.2872, 0.8704).

TC-CLIN-CPROP-006 CLOSED-FORM ANCHOR — odds ratio

Tests UR(s)	UR-CLIN-162
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv`
Rationale	OR = (15*70)/(85*30) = 0.4118; Woolf-log interval.
Pass criterion	Odds ratio 0.4118 (95% CI 0.2053, 0.8258).

TC-CLIN-CPROP-007 Uncorrected chi-square option

Tests UR(s)	UR-CLIN-163
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv --no-correction`
Rationale	Confirms the flag reaches chisq.test.
Pass criterion	X2 = 6.4516, df = 1, p = 0.01109.

TC-CLIN-CPROP-008 Reference group flips the odds ratio

Tests UR(s)	UR-CLIN-164
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv --reference treated`
Rationale	Swapping the reference inverts the OR.
Pass criterion	Odds ratio 2.4286 (= 1/0.4118), 95% CI (1.2110, 4.8703).

TC-CLIN-CPROP-009 Event level auto-detected

Tests UR(s)	UR-CLIN-165
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv`
Rationale	Reduces friction for common yes/no coding.

Pass criterion	Output reports event = 'yes'.

TC-CLIN-CPROP-010 Missing required column rejected

Tests UR(s)	UR-CLIN-166
Command / action	`jrc_clinical_compare_props means_sleep_student1908.csv`
Rationale	Fail-closed.
Pass criterion	Non-zero exit; 'Missing column(s)'.

TC-CLIN-CPROP-011 Duplicate id rejected

Tests UR(s)	UR-CLIN-167
Command / action	`jrc_clinical_compare_props props_dup_id.csv`
Rationale	Each subject once.
Pass criterion	Non-zero exit; 'Duplicate id(s) found'.

TC-CLIN-CPROP-012 Single group rejected

Tests UR(s)	UR-CLIN-168
Command / action	`jrc_clinical_compare_props props_one_group.csv`
Rationale	A comparison needs two groups.
Pass criterion	Non-zero exit; 'At least two groups'.

TC-CLIN-CPROP-013 Unknown --event rejected

Tests UR(s)	UR-CLIN-169
Command / action	`jrc_clinical_compare_props props_2x2_constructed.csv --event maybe`
Rationale	Fail-closed on a mistyped event label.
Pass criterion	Non-zero exit; 'not an outcome level'.

TC-CLIN-CPROP-014 Direct execution without RENV_PATHS_ROOT blocked

Tests UR(s)	UR-CLIN-170
Command / action	`Rscript repos/clinical/R/jrc_clinical_compare_props.R (no RENV_PATHS_ROOT)`
Rationale	The environment guard.
Pass criterion	Non-zero exit; 'RENV_PATHS_ROOT'.

Test Cases — jrc_clinical_ancova

TC-CLIN-ANCOVA-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-171
Command / action	`jrc_clinical_ancova ancova_anorexia.csv --covariates prewt`
Rationale	Confirms the script fits and completes.
Pass criterion	Exit 0; 'ANCOVA'; 72 subjects.

TC-CLIN-ANCOVA-002 PUBLISHED ANCHOR — anorexia Type III group test

Tests UR(s)	UR-CLIN-172
Command / action	`jrc_clinical_ancova ancova_anorexia.csv --covariates prewt`
Rationale	Anchored to the anorexia dataset (Venables & Ripley, MASS), the canonical ANCOVA example: Postwt ~ Prewt + Treat. Type III joint tests come from the validated emmeans package.
Pass criterion	group: F = 7.8680, df = 2, 68, p = 0.0008.

TC-CLIN-ANCOVA-003 Type III covariate test

Tests UR(s)	UR-CLIN-173
Command / action	`jrc_clinical_ancova ancova_anorexia.csv --covariates prewt`
Rationale	The covariate's adjusted contribution.
Pass criterion	prewt: F = 7.2660, df = 1, 68, p = 0.00885.

TC-CLIN-ANCOVA-004 PUBLISHED ANCHOR — adjusted least-squares means

Tests UR(s)	UR-CLIN-174
Command / action	`jrc_clinical_ancova ancova_anorexia.csv --covariates prewt`
Rationale	The adjusted means are the headline ANCOVA output.
Pass criterion	CBT 85.5743, Cont 81.4773, FT 90.1374.

TC-CLIN-ANCOVA-005 Tukey-adjusted pairwise contrast

Tests UR(s)	UR-CLIN-175
Command / action	`jrc_clinical_ancova ancova_anorexia.csv --covariates prewt`
Rationale	Which specific groups differ, after adjustment.
Pass criterion	Cont - FT diff -8.6601, p = 0.0005484.

TC-CLIN-ANCOVA-006 Covariate coefficient

Tests UR(s)	UR-CLIN-176
Command / action	`jrc_clinical_ancova ancova_anorexia.csv --covariates prewt`
Rationale	The per-unit covariate effect.
Pass criterion	prewt coef +0.4345 (SE 0.1612).

TC-CLIN-ANCOVA-007 Confidence level narrows the adjusted-mean CI

Tests UR(s)	UR-CLIN-177
Command / action	`jrc_clinical_ancova ancova_anorexia.csv --covariates prewt --conf 0.90`
Rationale	A lower level gives a tighter interval.
Pass criterion	CBT 90% CI (83.4121, 87.7365), inside the 95% CI.

TC-CLIN-ANCOVA-008 Missing required column rejected

Tests UR(s)	UR-CLIN-178
Command / action	`jrc_clinical_ancova means_sleep_student1908.csv --covariates prewt`
Rationale	Fail-closed.
Pass criterion	Non-zero exit; 'Missing column(s)'.

TC-CLIN-ANCOVA-009 --covariates required

Tests UR(s)	UR-CLIN-179
Command / action	`jrc_clinical_ancova ancova_anorexia.csv`
Rationale	ANCOVA needs a covariate.

Pass criterion	Non-zero exit; '--covariates is required'.

TC-CLIN-ANCOVA-010 Missing covariate column rejected

Tests UR(s)	UR-CLIN-180
Command / action	`jrc_clinical_ancova ancova_anorexia.csv --covariates nosuchcol`
Rationale	Fail-closed.
Pass criterion	Non-zero exit; 'Missing column(s)'.

TC-CLIN-ANCOVA-011 Duplicate id rejected

Tests UR(s)	UR-CLIN-181
Command / action	`jrc_clinical_ancova ancova_dup_id.csv --covariates prewt`
Rationale	Each subject once.
Pass criterion	Non-zero exit; 'Duplicate id(s) found'.

TC-CLIN-ANCOVA-012 Single group rejected

Tests UR(s)	UR-CLIN-182
Command / action	`jrc_clinical_ancova ancova_one_group.csv --covariates prewt`
Rationale	A comparison needs two groups.
Pass criterion	Non-zero exit; 'at least two levels'.

TC-CLIN-ANCOVA-013 Direct execution without RENV_PATHS_ROOT blocked

Tests UR(s)	UR-CLIN-183
Command / action	`Rscript repos/clinical/R/jrc_clinical_ancova.R (no RENV_PATHS_ROOT)`
Rationale	The environment guard.
Pass criterion	Non-zero exit; 'RENV_PATHS_ROOT'.

Test Cases — jrc_clinical_logistic

TC-CLIN-LOGIT-001 Valid inputs — exit 0

Tests UR(s)	UR-CLIN-184
Command / action	`jrc_clinical_logistic logistic_birthwt.csv --predictors age,lwt,smoke`
Rationale	Confirms the model fits and completes.
Pass criterion	Exit 0; 'Logistic regression'; 189 subjects.

TC-CLIN-LOGIT-002 PUBLISHED ANCHOR — birthwt smoke odds ratio

Tests UR(s)	UR-CLIN-185
Command / action	`jrc_clinical_logistic logistic_birthwt.csv --predictors age,lwt,smoke`
Rationale	Anchored to the Hosmer & Lemeshow low-birth-weight data (MASS 'birthwt'), the canonical logistic-regression example: low ~ age + lwt + smoke.
Pass criterion	smoke OR = 1.9557 (95% CI 1.0326, 3.7042).

TC-CLIN-LOGIT-003 Continuous predictor odds ratio

Tests UR(s)	UR-CLIN-186
Command / action	`jrc_clinical_logistic logistic_birthwt.csv --predictors age,lwt,smoke`
Rationale	Mother's weight is protective.
Pass criterion	lwt OR = 0.9879 (95% CI 0.9761, 0.9999).

TC-CLIN-LOGIT-004 Non-significant predictor reported

Tests UR(s)	UR-CLIN-187
Command / action	`jrc_clinical_logistic logistic_birthwt.csv --predictors age,lwt,smoke`
Rationale	A null result must be shown, not hidden.
Pass criterion	age OR = 0.9618, p = 0.2334.

TC-CLIN-LOGIT-005 Global likelihood-ratio test

Tests UR(s)	UR-CLIN-188
Command / action	`jrc_clinical_logistic logistic_birthwt.csv --predictors age,lwt,smoke`
Rationale	Overall model significance.
Pass criterion	chisq = 11.7926, df = 3, p = 0.008128.

TC-CLIN-LOGIT-006 Model fit (AIC)

Tests UR(s)	UR-CLIN-189
Command / action	`jrc_clinical_logistic logistic_birthwt.csv --predictors age,lwt,smoke`
Rationale	A model-comparison statistic.
Pass criterion	AIC = 230.88.

TC-CLIN-LOGIT-007 In-sample model AUC

Tests UR(s)	UR-CLIN-190
Command / action	`jrc_clinical_logistic logistic_birthwt.csv --predictors age,lwt,smoke`
Rationale	Reuses the tie-aware Mann-Whitney kernel validated for jrc_clinical_dx_roc.
Pass criterion	Model AUC (in-sample) = 0.6531.

TC-CLIN-LOGIT-008 Confidence level narrows the OR CI

Tests UR(s)	UR-CLIN-191
Command / action	`jrc_clinical_logistic logistic_birthwt.csv --predictors age,lwt,smoke --conf 0.90`
Rationale	A lower level gives a tighter interval.

Pass criterion	smoke OR CI (1.1442, 3.3427) at 90%, inside the 95% CI.

TC-CLIN-LOGIT-009 Missing required column rejected

Tests UR(s)	UR-CLIN-192
Command / action	`jrc_clinical_logistic means_sleep_student1908.csv --predictors value`
Rationale	Fail-closed.
Pass criterion	Non-zero exit; 'Missing column(s)'.

TC-CLIN-LOGIT-010 --predictors required

Tests UR(s)	UR-CLIN-193
Command / action	`jrc_clinical_logistic logistic_birthwt.csv`
Rationale	A model needs a predictor.
Pass criterion	Non-zero exit; '--predictors is required'.

TC-CLIN-LOGIT-011 Non-binary outcome rejected

Tests UR(s)	UR-CLIN-194
Command / action	`jrc_clinical_logistic log_three_outcomes.csv --predictors age`
Rationale	Logistic regression needs a binary outcome.
Pass criterion	Non-zero exit; 'exactly two levels'.

TC-CLIN-LOGIT-012 Duplicate id rejected

Tests UR(s)	UR-CLIN-195
Command / action	`jrc_clinical_logistic log_dup_id.csv --predictors age`
Rationale	Each subject once.
Pass criterion	Non-zero exit; 'Duplicate id(s) found'.

TC-CLIN-LOGIT-013 Direct execution without RENV_PATHS_ROOT blocked

Tests UR(s)	UR-CLIN-196
Command / action	`Rscript repos/clinical/R/jrc_clinical_logistic.R (no RENV_PATHS_ROOT)`
Rationale	The environment guard.
Pass criterion	Non-zero exit; 'RENV_PATHS_ROOT'.

Requirements Traceability Matrix

Maps each test case to the user requirement(s) it covers.

Test Case ID	User Requirement(s)
TC-CLIN-MEANS-001	UR-CLIN-001
TC-CLIN-MEANS-002	UR-CLIN-002
TC-CLIN-MEANS-003	UR-CLIN-003
TC-CLIN-MEANS-004	UR-CLIN-004
TC-CLIN-MEANS-005	UR-CLIN-005
TC-CLIN-MEANS-006	UR-CLIN-006
TC-CLIN-MEANS-007	UR-CLIN-007
TC-CLIN-MEANS-008	UR-CLIN-008
TC-CLIN-MEANS-009	UR-CLIN-009
TC-CLIN-MEANS-010	UR-CLIN-010
TC-CLIN-MEANS-011	UR-CLIN-011
TC-CLIN-MEANS-012	UR-CLIN-012
TC-CLIN-MEANS-013	UR-CLIN-013
TC-CLIN-MEANS-014	UR-CLIN-014
TC-CLIN-PROPS-001	UR-CLIN-015
TC-CLIN-PROPS-002	UR-CLIN-016
TC-CLIN-PROPS-003	UR-CLIN-017
TC-CLIN-PROPS-004	UR-CLIN-018
TC-CLIN-PROPS-005	UR-CLIN-019
TC-CLIN-PROPS-006	UR-CLIN-020
TC-CLIN-PROPS-007	UR-CLIN-021
TC-CLIN-PROPS-008	UR-CLIN-022
TC-CLIN-PROPS-009	UR-CLIN-023
TC-CLIN-PROPS-010	UR-CLIN-024
TC-CLIN-PROPS-011	UR-CLIN-025
TC-CLIN-SURV-001	UR-CLIN-026
TC-CLIN-SURV-002	UR-CLIN-027
TC-CLIN-SURV-003	UR-CLIN-028
TC-CLIN-SURV-004	UR-CLIN-029
TC-CLIN-SURV-005	UR-CLIN-030
TC-CLIN-SURV-006	UR-CLIN-031
TC-CLIN-SURV-007	UR-CLIN-032
TC-CLIN-SURV-008	UR-CLIN-033
TC-CLIN-SURV-009	UR-CLIN-034
TC-CLIN-SURV-010	UR-CLIN-035
TC-CLIN-SURV-011	UR-CLIN-036
TC-CLIN-DXACC-001	UR-CLIN-037
TC-CLIN-DXACC-002	UR-CLIN-038
TC-CLIN-DXACC-003	UR-CLIN-039
TC-CLIN-DXACC-004	UR-CLIN-040
TC-CLIN-DXACC-005	UR-CLIN-041
TC-CLIN-DXACC-006	UR-CLIN-042
TC-CLIN-DXACC-007	UR-CLIN-043
TC-CLIN-DXACC-008	UR-CLIN-044
TC-CLIN-DXACC-009	UR-CLIN-045
TC-CLIN-DXACC-010	UR-CLIN-046
TC-CLIN-DXACC-011	UR-CLIN-047
TC-CLIN-DXACC-012	UR-CLIN-048
TC-CLIN-DXACC-013	UR-CLIN-049

TC-CLIN-DXACC-014	UR-CLIN-050
TC-CLIN-DXACC-015	UR-CLIN-051
TC-CLIN-DXACC-016	UR-CLIN-052
TC-CLIN-DXACC-017	UR-CLIN-053
TC-CLIN-DXACC-018	UR-CLIN-054
TC-CLIN-DXACC-019	UR-CLIN-055
TC-CLIN-DXACC-020	UR-CLIN-056
TC-CLIN-DXSS-001	UR-CLIN-057
TC-CLIN-DXSS-002	UR-CLIN-058
TC-CLIN-DXSS-003	UR-CLIN-059
TC-CLIN-DXSS-004	UR-CLIN-060
TC-CLIN-DXSS-005	UR-CLIN-061
TC-CLIN-DXSS-006	UR-CLIN-062
TC-CLIN-DXSS-007	UR-CLIN-063
TC-CLIN-DXSS-008	UR-CLIN-064
TC-CLIN-DXSS-009	UR-CLIN-065
TC-CLIN-DXSS-010	UR-CLIN-066
TC-CLIN-DXSS-011	UR-CLIN-067
TC-CLIN-DXSS-012	UR-CLIN-068
TC-CLIN-DXSS-013	UR-CLIN-069
TC-CLIN-DXSS-014	UR-CLIN-070
TC-CLIN-DXROC-001	UR-CLIN-071
TC-CLIN-DXROC-002	UR-CLIN-072
TC-CLIN-DXROC-003	UR-CLIN-073
TC-CLIN-DXROC-004	UR-CLIN-074
TC-CLIN-DXROC-005	UR-CLIN-075
TC-CLIN-DXROC-006	UR-CLIN-076
TC-CLIN-DXROC-007	UR-CLIN-077
TC-CLIN-DXROC-008	UR-CLIN-078
TC-CLIN-DXROC-009	UR-CLIN-079
TC-CLIN-DXROC-010	UR-CLIN-080
TC-CLIN-DXROC-011	UR-CLIN-081
TC-CLIN-DXROC-012	UR-CLIN-082
TC-CLIN-DXROC-013	UR-CLIN-083
TC-CLIN-DXROC-014	UR-CLIN-084
TC-CLIN-DXROC-015	UR-CLIN-085
TC-CLIN-DXROC-016	UR-CLIN-086
TC-CLIN-DXROC-017	UR-CLIN-087
TC-CLIN-KM-001	UR-CLIN-088
TC-CLIN-KM-002	UR-CLIN-089
TC-CLIN-KM-003	UR-CLIN-090
TC-CLIN-KM-004	UR-CLIN-091
TC-CLIN-KM-005	UR-CLIN-092
TC-CLIN-KM-006	UR-CLIN-093
TC-CLIN-KM-007	UR-CLIN-094
TC-CLIN-KM-008	UR-CLIN-095
TC-CLIN-KM-009	UR-CLIN-096
TC-CLIN-KM-010	UR-CLIN-097
TC-CLIN-KM-011	UR-CLIN-098
TC-CLIN-KM-012	UR-CLIN-099
TC-CLIN-KM-013	UR-CLIN-100
TC-CLIN-KM-014	UR-CLIN-101
TC-CLIN-KM-015	UR-CLIN-102
TC-CLIN-KM-016	UR-CLIN-103
TC-CLIN-KM-017	UR-CLIN-104
TC-CLIN-KM-018	UR-CLIN-105

TC-CLIN-KM-019	UR-CLIN-106
TC-CLIN-COX-001	UR-CLIN-107
TC-CLIN-COX-002	UR-CLIN-108
TC-CLIN-COX-003	UR-CLIN-109
TC-CLIN-COX-004	UR-CLIN-110
TC-CLIN-COX-005	UR-CLIN-111
TC-CLIN-COX-006	UR-CLIN-112
TC-CLIN-COX-007	UR-CLIN-113
TC-CLIN-COX-008	UR-CLIN-114
TC-CLIN-COX-009	UR-CLIN-115
TC-CLIN-COX-010	UR-CLIN-116
TC-CLIN-COX-011	UR-CLIN-117
TC-CLIN-COX-012	UR-CLIN-118
TC-CLIN-COX-013	UR-CLIN-119
TC-CLIN-COX-014	UR-CLIN-120
TC-CLIN-COX-015	UR-CLIN-121
TC-CLIN-COX-016	UR-CLIN-122
TC-CLIN-COX-017	UR-CLIN-123
TC-CLIN-DXCMP-001	UR-CLIN-124
TC-CLIN-DXCMP-002	UR-CLIN-125
TC-CLIN-DXCMP-003	UR-CLIN-126
TC-CLIN-DXCMP-004	UR-CLIN-127
TC-CLIN-DXCMP-005	UR-CLIN-128
TC-CLIN-DXCMP-006	UR-CLIN-129
TC-CLIN-DXCMP-007	UR-CLIN-130
TC-CLIN-DXCMP-008	UR-CLIN-131
TC-CLIN-DXCMP-009	UR-CLIN-132
TC-CLIN-DXCMP-010	UR-CLIN-133
TC-CLIN-DXCMP-011	UR-CLIN-134
TC-CLIN-DXCMP-012	UR-CLIN-135
TC-CLIN-DXCMP-013	UR-CLIN-136
TC-CLIN-DXCMP-014	UR-CLIN-137
TC-CLIN-DXCMP-015	UR-CLIN-138
TC-CLIN-DXCMP-016	UR-CLIN-139
TC-CLIN-DXCMP-017	UR-CLIN-140
TC-CLIN-DXCMP-018	UR-CLIN-141
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TC-CLIN-CMEAN-002	UR-CLIN-143
TC-CLIN-CMEAN-003	UR-CLIN-144
TC-CLIN-CMEAN-004	UR-CLIN-145
TC-CLIN-CMEAN-005	UR-CLIN-146
TC-CLIN-CMEAN-006	UR-CLIN-147
TC-CLIN-CMEAN-007	UR-CLIN-148
TC-CLIN-CMEAN-008	UR-CLIN-149
TC-CLIN-CMEAN-009	UR-CLIN-150
TC-CLIN-CMEAN-010	UR-CLIN-151
TC-CLIN-CMEAN-011	UR-CLIN-152
TC-CLIN-CMEAN-012	UR-CLIN-153
TC-CLIN-CMEAN-013	UR-CLIN-154
TC-CLIN-CMEAN-014	UR-CLIN-155
TC-CLIN-CMEAN-015	UR-CLIN-156
TC-CLIN-CPROP-001	UR-CLIN-157
TC-CLIN-CPROP-002	UR-CLIN-158
TC-CLIN-CPROP-003	UR-CLIN-159
TC-CLIN-CPROP-004	UR-CLIN-160
TC-CLIN-CPROP-005	UR-CLIN-161

TC-CLIN-CPROP-006	UR-CLIN-162
TC-CLIN-CPROP-007	UR-CLIN-163
TC-CLIN-CPROP-008	UR-CLIN-164
TC-CLIN-CPROP-009	UR-CLIN-165
TC-CLIN-CPROP-010	UR-CLIN-166
TC-CLIN-CPROP-011	UR-CLIN-167
TC-CLIN-CPROP-012	UR-CLIN-168
TC-CLIN-CPROP-013	UR-CLIN-169
TC-CLIN-CPROP-014	UR-CLIN-170
TC-CLIN-ANCOVA-001	UR-CLIN-171
TC-CLIN-ANCOVA-002	UR-CLIN-172
TC-CLIN-ANCOVA-003	UR-CLIN-173
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TC-CLIN-LOGIT-006	UR-CLIN-189
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TC-CLIN-LOGIT-011	UR-CLIN-194
TC-CLIN-LOGIT-012	UR-CLIN-195
TC-CLIN-LOGIT-013	UR-CLIN-196